



US012413163B2

(12) **United States Patent**
Sanchez

(10) **Patent No.:** **US 12,413,163 B2**
(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **HARVESTING ENERGY FOR A SMART RING VIA PIEZOELECTRIC CHARGING**

(71) Applicant: **QUANATA, LLC**, San Francisco, CA (US)

(72) Inventor: **Kenneth Jason Sanchez**, San Francisco, CA (US)

(73) Assignee: **QUANATA, LLC**, San Francisco, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/594,294**

(22) Filed: **Mar. 4, 2024**

(65) **Prior Publication Data**

US 2024/0250622 A1 Jul. 25, 2024

Related U.S. Application Data

(63) Continuation of application No. 18/187,745, filed on Mar. 22, 2023, now Pat. No. 11,923,791, which is a (Continued)

(51) **Int. Cl.**
H01M 10/44 (2006.01)
A44C 9/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H02N 2/188** (2013.01); **A44C 9/0053** (2013.01); **H02J 7/007** (2013.01); **H02J 50/001** (2020.01);
(Continued)

(58) **Field of Classification Search**
CPC H02J 7/007; H02J 7/0042; H02J 50/001; H02J 50/005; H02J 50/10; H02N 2/188;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

513,344 A 1/1894 Smith
2,737,192 A 3/1956 Bieler
(Continued)

FOREIGN PATENT DOCUMENTS

CN 103109462 A 5/2013
CN 104799509 7/2015
(Continued)

OTHER PUBLICATIONS

“How to find your ideal bedtime with the Oura app”, available online at <<https://web.archive.org/web/20191206205332/https://ouraring.com/how-find-your-ideal-bedtime-with-the-oura-app/>>, 2019, 8 pages. 2019.

(Continued)

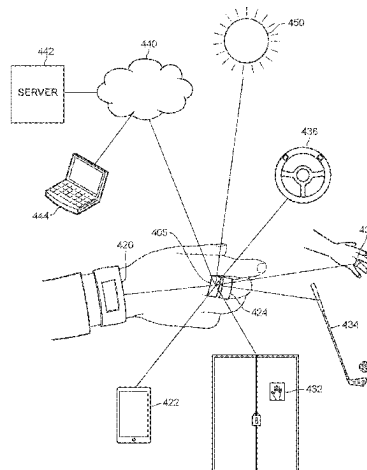
Primary Examiner — Edward Tso

(74) *Attorney, Agent, or Firm* — BRYAN CAVE
LEIGHTON PAISNER LLP

(57) **ABSTRACT**

A smart ring includes a housing configured to be worn by a user. The smart ring can further include a power source disposed within the housing. The smart ring also can include a charging circuit disposed within the housing. The charging circuit can include a harvesting element configured to charge the power source. The harvesting element can be configured to convert mechanical energy to electrical energy. The smart ring also can include a device at least partially disposed within the housing. The device can be configured to draw energy from the power source. Other embodiments are disclosed.

20 Claims, 7 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/926,270, filed on
Jul. 10, 2020, now Pat. No. 11,637,511.

(51) Int. Cl.

H01M 10/46 (2006.01)
H02J 7/00 (2006.01)
H02J 50/00 (2016.01)
H02J 50/10 (2016.01)
H02K 11/049 (2016.01)
H02K 11/05 (2016.01)
H02K 35/02 (2006.01)
H02N 2/18 (2006.01)
H10N 30/30 (2023.01)

(52) U.S. Cl.

CPC **H02J 50/10** (2016.02); **H02K 11/049**
(2016.01); **H02K 11/05** (2016.01); **H02K**
35/02 (2013.01); **H02N 2/181** (2013.01);
H10N 30/306 (2023.02)

(58) Field of Classification Search

CPC H02N 2/181; H02N 30/306; H02K 11/05;
H02K 35/02; H02K 11/049; A44C 9/0053
USPC 320/101, 107, 108, 114, 115
See application file for complete search history.

(56) References Cited**U.S. PATENT DOCUMENTS**

3,792,192	A	2/1974	Plate	9,362,775	B1	6/2016	Jacobs
4,012,629	A	3/1977	Simms	9,420,260	B2	8/2016	McGregor et al.
4,382,112	A	5/1983	Betts et al.	9,440,657	B1	9/2016	Fields et al.
4,684,687	A	8/1987	Breach et al.	9,477,146	B2	10/2016	Xu et al.
4,830,014	A	5/1989	Goodman et al.	9,509,170	B2	11/2016	Wu
4,880,304	A	11/1989	Jaeb et al.	9,628,707	B2	4/2017	Blum et al.
5,135,220	A	8/1992	Baldoni	9,660,488	B2	5/2017	Breedvelt-Schouten et al.
6,097,480	A	8/2000	Kaplan	9,696,690	B2	7/2017	Nguyen et al.
6,154,658	A	11/2000	Caci	9,711,060	B1	7/2017	Lusted et al.
6,201,698	B1	3/2001	Hunter	9,711,993	B2	7/2017	Kim
6,560,993	B1	5/2003	Bosque et al.	9,717,949	B1	8/2017	Tran et al.
6,608,562	B1	8/2003	Kimura et al.	9,756,301	B2	9/2017	Li et al.
6,699,199	B2	3/2004	Asada et al.	9,841,331	B2	12/2017	Wood et al.
6,745,061	B1	6/2004	Hicks et al.	9,847,020	B2	12/2017	Davis
6,792,044	B2	9/2004	Peng et al.	9,861,314	B2	1/2018	Haverinen et al.
6,800,693	B2	10/2004	Nishihara et al.	9,908,530	B1	3/2018	Fields et al.
6,803,391	B2	10/2004	Paglia et al.	9,931,976	B1	4/2018	Terwilliger et al.
6,805,140	B2	10/2004	Velez, Jr. et al.	9,955,286	B2	4/2018	Segal
6,894,628	B2	5/2005	Marpe et al.	9,956,963	B2	5/2018	Vijaya Kumar et al.
6,909,753	B2	6/2005	Meehan et al.	9,965,761	B2	5/2018	Elangovan et al.
6,959,116	B2	10/2005	Sezer et al.	10,007,355	B2	6/2018	Schorsch et al.
7,013,674	B2	3/2006	Kretchmer	10,085,695	B2	10/2018	Ouwerkerk et al.
7,136,532	B2	11/2006	VanDer	10,099,608	B2	10/2018	Cuddihy et al.
7,190,986	B1	3/2007	Hannula et al.	10,102,510	B2	10/2018	Yau et al.
7,227,894	B2	6/2007	Lin et al.	10,137,777	B2	11/2018	Lu et al.
7,275,172	B2	9/2007	Janzen	10,139,859	B2	11/2018	Von Badinski et al.
7,286,710	B2	10/2007	Marpe et al.	10,281,953	B2	5/2019	Von Badinski
7,468,036	B1	12/2008	Rulkov et al.	10,303,867	B2	5/2019	Schröder
7,500,746	B1	3/2009	Howell et al.	10,315,557	B2	6/2019	Terwilliger et al.
7,519,229	B2	4/2009	Wallace et al.	10,317,940	B2	6/2019	Eim et al.
7,689,437	B1	3/2010	Teller et al.	10,345,506	B1	7/2019	Lyu
7,872,444	B2	1/2011	Hamilton et al.	10,359,846	B2	7/2019	Priyantha et al.
8,031,172	B2	10/2011	Kruse et al.	10,366,220	B2	7/2019	Shapiro et al.
8,075,484	B2	12/2011	Moore-Ede	10,377,386	B2	8/2019	Richmond
8,345,752	B2	1/2013	Lee et al.	10,396,584	B2	8/2019	Madau et al.
8,446,275	B2	5/2013	Utter, II	10,409,327	B2	9/2019	Stotler
8,554,297	B2	10/2013	Moon et al.	10,444,834	B2	10/2019	Vescovi et al.
8,570,273	B1	10/2013	Smith	10,463,141	B2	11/2019	Fitzgerald et al.
8,700,111	B2	4/2014	Leboeuf et al.	10,528,989	B1	1/2020	Irey
8,954,135	B2	2/2015	Yuen et al.	10,564,628	B2	2/2020	Hargovan et al.
9,218,058	B2	12/2015	Bress et al.	10,629,175	B2	4/2020	Yan et al.
9,248,839	B1	2/2016	Tan	10,664,842	B1	5/2020	Bermudez et al.
9,248,938	B2	2/2016	Hopps	10,693,872	B1	6/2020	Larson et al.
9,265,310	B2	2/2016	Lam	10,703,204	B2	7/2020	Hassan et al.
				10,709,339	B1	7/2020	Lusted
				10,745,032	B2	8/2020	Scheggi
				10,762,183	B1	9/2020	Charan et al.
				10,768,666	B2	9/2020	Von Badinski et al.
				10,838,499	B2	11/2020	Wang
				10,842,429	B2	11/2020	Kinnunen et al.
				10,849,557	B2	12/2020	Keating et al.
				10,893,833	B2	1/2021	Haverinen et al.
				10,944,745	B2	3/2021	Kursun et al.
				11,227,060	B1	1/2022	John et al.
				11,265,635	B2	3/2022	Shankar
				11,312,299	B1	4/2022	Assam
				11,479,258	B1	10/2022	Sanchez
				11,599,147	B2	3/2023	Von Badinski et al.
				11,637,511	B2 *	4/2023	Sanchez H02N 2/18 320/107
				11,868,178	B2	1/2024	Von Badinski et al.
				11,868,179	B2	1/2024	Von Badinski et al.
				11,894,704	B2	2/2024	Sanchez
				11,909,238	B1	2/2024	Sanchez
				11,923,791	B2 *	3/2024	Sanchez H02J 7/007
				11,984,742	B2	5/2024	Sanchez
				12,191,692	B2	1/2025	Sanchez
				12,211,467	B2	1/2025	Sanchez
				12,237,700	B2	2/2025	Sanchez et al.
				2002/0042464	A1	4/2002	Barclay et al.
				2002/0091568	A1	7/2002	Kraft et al.
				2002/0121831	A1	9/2002	Egawa et al.
				2003/0077064	A1	4/2003	Katayama
				2003/0142065	A1	7/2003	Pahlavan
				2004/0102551	A1	5/2004	Sato et al.
				2004/0118592	A1	6/2004	Pehlert
				2004/0145256	A1	7/2004	Miekka et al.
				2004/0160635	A1	8/2004	Ikeda et al.
				2004/0200235	A1	10/2004	Kretchmer

(56)

References Cited

U.S. PATENT DOCUMENTS

2005/0012648	A1	1/2005	Marpe et al.	2016/0236692	A1	8/2016	Kleen et al.
2005/0030205	A1	2/2005	Konoshima et al.	2016/0292563	A1	10/2016	Park
2005/0054941	A1	3/2005	Ting et al.	2016/0317060	A1	11/2016	Connor
2005/0062454	A1	3/2005	Raghunath et al.	2016/0334901	A1	11/2016	Rihn
2005/0133248	A1	6/2005	Easter	2016/0336758	A1	11/2016	Breedvelt-Schouten et al.
2005/0162523	A1	7/2005	Darrell et al.	2016/0350581	A1	12/2016	Manuel et al.
2005/0185060	A1	8/2005	Neven, Sr.	2016/0361032	A1	12/2016	Carter et al.
2005/0185843	A1	8/2005	Kudoh	2017/0010677	A1	1/2017	Roh et al.
2005/0185844	A1	8/2005	Ono et al.	2017/0012925	A1	1/2017	Tekin et al.
2005/0230596	A1	10/2005	Howell et al.	2017/0024008	A1	1/2017	Kienzle et al.
2006/0002607	A1	1/2006	Boncyk et al.	2017/0026790	A1	1/2017	Flitsch et al.
2006/0069681	A1	3/2006	Lauper	2017/0042477	A1	2/2017	Haverinen et al.
2006/0080286	A1	4/2006	Svendsen	2017/0053461	A1	2/2017	Pal et al.
2006/0085477	A1	4/2006	Phillips et al.	2017/0057492	A1	3/2017	Eddington et al.
2006/0089792	A1	4/2006	Manber et al.	2017/0070078	A1	3/2017	Hwang et al.
2006/0211924	A1	9/2006	Dalke et al.	2017/0075701	A1	3/2017	Ricci et al.
2006/0250043	A1	11/2006	Chung	2017/0080952	A1	3/2017	Gupta et al.
2006/0271593	A1	11/2006	De Mes et al.	2017/0090475	A1	3/2017	Choi et al.
2007/0149222	A1	6/2007	Hodko et al.	2017/0109512	A1	4/2017	Bower et al.
2007/0159522	A1	7/2007	Neven	2017/0129335	A1	5/2017	Lu et al.
2007/0188626	A1	8/2007	Squilla et al.	2017/0131772	A1	5/2017	Choi
2007/0200713	A1	8/2007	Weber et al.	2017/0190121	A1	7/2017	Aggarwal et al.
2007/0223826	A1	9/2007	Ridge et al.	2017/0192530	A1	7/2017	Lee
2008/0068559	A1	3/2008	Howell et al.	2017/0242428	A1	8/2017	Pal et al.
2008/0136587	A1	6/2008	Orr	2017/0251967	A1	9/2017	Premasukh
2008/0174676	A1	7/2008	Squilla et al.	2017/0336964	A1	11/2017	Kim
2008/0218684	A1	9/2008	Howell et al.	2017/0346635	A1	11/2017	Gummesson et al.
2008/0275309	A1	11/2008	Stivorac et al.	2017/0347895	A1	12/2017	Wei et al.
2009/0056703	A1	3/2009	Mills et al.	2017/0374074	A1	12/2017	Stuntebeck
2010/0219989	A1	9/2010	Asami et al.	2018/0025351	A1	1/2018	Chen et al.
2011/0007035	A1	1/2011	Shai	2018/0025430	A1	1/2018	Perl et al.
2011/0080339	A1	4/2011	Sun et al.	2018/0032126	A1	2/2018	Liu
2011/0224875	A1	9/2011	Cuddihy et al.	2018/0037228	A1	2/2018	Biondo et al.
2012/0016245	A1	1/2012	Niwa et al.	2018/0039303	A1	2/2018	Hashimoto et al.
2012/0130203	A1	5/2012	Stergiou et al.	2018/0052428	A1	2/2018	Abramov
2012/0184367	A1	7/2012	Parrott et al.	2018/0054513	A1	2/2018	Ma
2012/0218184	A1	8/2012	Wissmar	2018/0068105	A1*	3/2018	Shapiro G06F 21/602
2012/0293107	A1	11/2012	Ajagbe	2018/0093606	A1	4/2018	Terwilliger et al.
2012/0317024	A1	12/2012	Rahman et al.	2018/0093610	A1	4/2018	Sun et al.
2013/0106603	A1	5/2013	Weast et al.	2018/0093672	A1	4/2018	Terwilliger et al.
2013/0211291	A1	8/2013	Tran	2018/0115797	A1	4/2018	Wexler et al.
2013/0335213	A1	12/2013	Sherony et al.	2018/0120891	A1	5/2018	Eim
2014/0107493	A1	4/2014	Yuen et al.	2018/0120892	A1	5/2018	Von Badinski et al.
2014/0118704	A1	5/2014	Duelli et al.	2018/0123629	A1	5/2018	Wetzig
2014/0120983	A1	5/2014	Lam	2018/0167200	A1	6/2018	High et al.
2014/0187160	A1	7/2014	Prencipe	2018/0174457	A1	6/2018	Taylor
2014/0218529	A1	8/2014	Mahmoud et al.	2018/0178712	A1	6/2018	Terwilliger et al.
2014/0238153	A1	8/2014	Wood et al.	2018/0229674	A1	8/2018	Heinrich et al.
2014/0240132	A1	8/2014	Bychkov	2018/0256027	A1	9/2018	Lacher
2014/0244009	A1	8/2014	Mestas	2018/0257668	A1	9/2018	Tonshal
2014/0274203	A1	9/2014	Ganong, III	2018/0262505	A1	9/2018	Ligatti
2014/0309849	A1	10/2014	Ricci	2018/0292901	A1	10/2018	Priyantha et al.
2014/0361934	A1	12/2014	Ely et al.	2018/0300467	A1	10/2018	Kwong et al.
2014/0361945	A1	12/2014	Misra et al.	2018/0322957	A1	11/2018	Dill et al.
2015/0003693	A1	1/2015	Baca et al.	2019/0004325	A1	1/2019	Connor
2015/0019266	A1	1/2015	Stempora	2019/0049267	A1	2/2019	Huang
2015/0028996	A1	1/2015	Agrafioti	2019/0083022	A1	3/2019	Huang
2015/0046996	A1	2/2015	Slaby et al.	2019/0131812	A1	5/2019	Lee et al.
2015/0062086	A1	3/2015	Nattukallingal	2019/0155104	A1	5/2019	Li et al.
2015/0065090	A1	3/2015	Yeh	2019/0155385	A1	5/2019	Lim et al.
2015/0098309	A1	4/2015	Adams	2019/0172289	A1	6/2019	O'Toole et al.
2015/0124096	A1	5/2015	Koravadi	2019/0191998	A1	6/2019	Heikenfeld et al.
2015/0126824	A1	5/2015	Leboeuf et al.	2019/0213429	A1	7/2019	Sicconi et al.
2015/0158499	A1	6/2015	Koravadi	2019/0230507	A1	7/2019	Li et al.
2015/0186092	A1	7/2015	Francis et al.	2019/0265868	A1	8/2019	Penilla et al.
2015/0220109	A1	8/2015	VonBadinski et al.	2019/0286805	A1	9/2019	Law et al.
2015/0277559	A1	10/2015	Vescovi et al.	2019/0287083	A1	9/2019	Wurmfeld et al.
2015/0338926	A1	11/2015	Park et al.	2019/0295440	A1	9/2019	Hadad
2015/0352953	A1	12/2015	Koravadi	2019/0298173	A1	10/2019	Lawrence et al.
2016/0028267	A1	1/2016	Lee et al.	2019/0298265	A1	10/2019	Keating
2016/0098530	A1	4/2016	Dill et al.	2019/0313967	A1	10/2019	Lee
2016/0189149	A1	6/2016	MacLaurin et al.	2019/0332140	A1	10/2019	Wang et al.
2016/0192407	A1	6/2016	Fyfe et al.	2019/0342329	A1	11/2019	Turgeman
2016/0207454	A1	7/2016	Cuddihy et al.	2019/0357834	A1	11/2019	Aarts et al.
2016/0226313	A1	8/2016	Okubo	2020/0001895	A1	1/2020	Scheggi
				2020/0005791	A1	1/2020	Rakshit et al.
				2020/0062276	A1	2/2020	Yuan et al.
				2020/0070840	A1	3/2020	Gunaratne
				2020/0142497	A1	5/2020	Zhu

(56)

References Cited**U.S. PATENT DOCUMENTS**

2020/0159896	A1	5/2020	Shapiro et al.
2020/0218238	A1	7/2020	Wang
2020/0356652	A1	11/2020	Yamaguchi et al.
2020/0391696	A1	12/2020	Kato et al.
2021/0019731	A1	1/2021	Rule et al.
2021/0029112	A1	1/2021	Palle et al.
2021/0058692	A1	2/2021	Shankar
2021/0197849	A1	7/2021	Tsuji
2021/0382684	A1	12/2021	Hachiya et al.
2022/0083149	A1	3/2022	Keller et al.

FOREIGN PATENT DOCUMENTS

CN	105960196	9/2016
CN	106360895	2/2017
CN	206213423	6/2017
CN	206333477	7/2017
CN	206371611	8/2017
CN	107139933	9/2017
CN	107260139	10/2017
CN	104157116	A 12/2017
CN	105006103	B 4/2018
CN	105841851	8/2018
CN	108900691	11/2018
CN	108926081	12/2018
DE	10201301233399	7/2013
DE	102015006677	11/2016
DE	102019116618	12/2020
EP	1223191	A1 7/2002
EP	1384752	A1 1/2004
EP	2281205	A1 2/2011
EP	2581856	4/2013
JP	200879676	4/2008
KR	20110012229	2/2011
KR	20170013067	2/2017
KR	1020170087113	7/2017
KR	101835991	B1 4/2018
WO	2001017421	3/2001
WO	2005114476	A1 12/2005
WO	2005124594	A1 12/2005
WO	2008008714	A1 1/2008
WO	2011132009	A2 4/2011
WO	2015077418	5/2015
WO	2017136940	8/2017
WO	2018000396	1/2018
WO	2018048563	A1 3/2018
WO	2018154341	8/2018
WO	2018164632	9/2018
WO	2018204811	11/2018
WO	2019082095	5/2019
WO	2019140528	7/2019
WO	2019180626	9/2019

OTHER PUBLICATIONS

“Vauxhall/Opel In-Car Wireless Charging”, retrieved from <<https://www.air-charge.com/aircharge-for-business/automotive/vauxhall-wireless-charging>>, Oct. 2019, 4 pages. 2019.

“Wireless charging for smart ring/pointing devices” available online at <http://www.humavox.com/smt_product/wireless-charging-for-smart-ringpointing-devices/>, Oct. 2019, 3 pages. 2019.

Adafruit, p. 1-2, available at: <https://adafruit.com/product/2806>, published Jun. 2019 (Year: 2019) 2019.

Adafruit.com, “RFID/NFC Smart Ring—Size 12—NTAG213,” accessed at <https://web.archive.org/web/20190605061438/https://www.adafruit.com/product/2806>, publication Jun. 5, 2019 Jun. 5, 2019.

ASU projection wearable: Live tomorrow today (world first launch @ CES 2016). (Dec. 2015). ASU Tech, YouTube. Retrieved from <https://www.youtube.com/watch?v=Wdb5O-D7Y0Y> 2015.

Brownell, L., “Low-cost wearables manufactured by hybrid 3D printing. Wyss Institute, Harvard,” Retrieved from <https://wyss.harvard.edu/news/low-cost-wearables-manufactured-by-hybrid-3d-printing/>, Sep. 6, 2017, pp. 11 2017.

harvard.edu/news/low-cost-wearables-manufactured-by-hybrid-3d-printing/, Sep. 6, 2017, pp. 11 2017.

Cetin, C., “Design, testing and implementation of a new authentication method using multiple devices,” Graduate Theses and Dissertations, University of South Florida Scholar Commons. Retrieved from <http://scholarcommons.usf.edu/etd/5660>, Jan. 2015, pp. 61 Jan. 2015.

Charles Q. Choi, “Low Battery? New Tech Lets You Wirelessly Share Power”, available online at <<https://www.livescience.com/54790-new-tech-enables-wireless-charging.html>>, May 19, 2016, 9 pages May 19, 2016.

Chen, X. A., et al., “Encore: 3D printed augmentation of everyday objects with printed-over, affixed and interlocked attachments,” Nov. 5, 2015, pp. 73-82 2015.

Chen, X. A., et al., “Reprise: A design tool for specifying, generating, and customizing 3D printable adaptations on everyday objects,” Oct. 16, 2016, pp. 29-39 Oct. 16, 2016.

E-Senses, “Personal vitamin D, sunlight and daylight coach”, available online at <<https://e-senses.com/>>, 2019, 5 pages 2019.

Google translation of KR20170087113A (Year: 2016) 2016.

Hipolite, W., “The 3D printed O Bluetooth Ring is one of the tiniest personal computers you will ever see,” 3DPrint.com. Retrieved from <https://3dprint.com/34627/o-bluetooth-ring-3d-printed/>, Jan. 2015, pp. 5 Jan. 2015.

https://en.wikipedia.org/w/index.php?title=Ring_size&oldid=891328817 2019.

Hussain Almossawi, “This smart ring aims to provide better lives for people with sickle cell disease”, retrieved from <<https://www.core77.com/projects/82131/This-Smart-Ring-Aims-to-Provide-Better-Lives-for-People-with-Sickle-Cell-Disease>>, 2021, 9 pages. 2021.

Je et al., “PokeRing: Notifications by poking around the finger”, Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems—CHI’18, 2018, paper 542, pp. 1-10 2018.

Katharine Schwab, “Orie, the ring that turns your finger into a phone, is here”, available online at <<https://www.fastcompany.com/90399237/orie-the-ring-that-turns-your-finger-into-a-phone-is-here>>, 2019, 4 pages 2019.

Laput et al., “Skin buttons: cheap, small, low-powered and clickable fixed-icon laser projectors”, UIST ’14: Proceedings of the 27th annual ACM symposium on User interface software and technology, Oct. 2014 pp. 389-394. 2014.

Magno et al., “Self-sustainable smart ring for long-term monitoring of blood oxygenation”, IEEE Access, 2019, pp. 115400-115408 2019.

Mahmud et al., Wearable technology for drug abuse detection: A survey of recent advancements, Smart Health, vol. 13, Aug. 2019, 100062 Aug. 2019.

Margaret, “The Orb: A Bluetooth headset that turns into a ring”, Gadgets, BornRich, Jun. 2013, available online at <<http://www.bornrich.com/the-orb-a-bluetooth-headset-that-turns-into-a-ring.html>>, Jun. 2013.

Mario, <https://www.smartringnews.com/posts/smart-ring-vs-smartwatch-which-is-the-best-fitness-and-activity-tracker> 2014.

Nassi et al., “Virtual breathalyzer”, Department of Software and Information Systems Engineering, Ben-Gurion University of the Negev, Israel, 2016, 10 pages 2016.

Neev Kiran, “SkinnySensor: Enabling Battery-Less Wearable Sensors via Intrabody Power Transfer”, Masters Theses 694, University of Massachusetts Amherst, 2018, 63 pages 2018.

Nerd-Fu, “Push present”, Delicious Juice Dot Com, Apr. 2015, available online at <<https://blog.deliciousjuice.com/2015/04/>> Apr. 2015.

Pablo E Suarez, “NXT Ring—Your Digital-self at Hand”, available online at <<https://www.youtube.com/watch?v=9w7uxDHs7NY>>, uploaded on Jun. 21, 2019, 2 pages Jun. 21, 2019.

Roumen et al., “NotiRing: A comparative study of notification channels for wearable interactive rings”, Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems—CHI’15, 2015, pp. 2497-2500 2015.

Sarah Jacobsson Purewal, “Ringly review: The smart ring that could be sexier”, available online at <<https://www.macworld.com/article/227133/ringly-review-the-smart-ring-that-could-be-sexier.html>>, 2016, 10 pages 2016.

(56)

References Cited

OTHER PUBLICATIONS

- Schwab, K., "This startup wants to kill passwords—and replace them with jewelry. Fast Company," Retrieved from <https://www.fastcompany.com/90254843/this-startup-wants-to-kill-passwords-and-replace-them-with-jewelry>, (Oct. 2018), pp. 7 Oct. 2018.
- Seung et al., "Nanopatterned Textile-Based Wearable Triboelectric Nanogenerator", ACS Nano, vol. 9, 2015, pp. 3501-3509 2015.
- Shane McGlaun, "Geek builds Bluetooth Smart Ring with OLED display", available online at <<https://www.slashgear.com/geek-builds-bluetooth-smart-ring-with-oled-display-02361383/>>, 2015, 6 pages 2015.
- Sperlazza, "We tested four sleep tracker apps and wearables: Here are the best ones", available online at <<https://www.bulletproof.com/sleep/tech/best-sleep-tracker-apps/>>, 2019, 18 pages 2019.
- Turunen, "Smart ring for stress control and self-understanding", available online at <<https://slowfinland.fi/en/smart-ring-for-stress-control-and-self-understanding/>>, 2017, 9 pages 2017.
- Woochit Tech (2017). New smart ring monitors UV exposure [video file]. Retrieved from <https://www.youtube.com/watch?v=4YvkioIZxjU>, 3 pages. 2017.
- Worgan et al., "Garment level power distribution for wearables using inductive power transfer," 9th International Conference on Human System Interactions (HSI), 2016, pp. 277-283 2016.
- Xiao et al., "LumiWatch: On-arm projected graphics and touch input," Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems—CHI'18, 2018, pp. 1-11 2018.
- Zhu et al., "Developing a driving fatigue detection system using physiological sensors," Proceedings of the 29th Australian Conference on Computer-Human Interaction—OZCHI'17, 2017, pp. 566-570 2017.
- Zhu, M., et al., "Fluidic fabric muscle sheets for wearable and soft robotics," Retrieved from <https://arxiv.org/pdf/1903.08253.pdf>, Mar. 2019, pp. 32 Mar. 2019.
- "The Oura App | Oura Ring", available online at <<https://web.archive.org/web/20191019192921/https://ouraring.com/introducing-the-new-oura-app/>>, 6 pages Oct. 29, 2019.
- "Get the Technical Specs of Oura Ring | Oura Ring", available online at <<https://web.archive.org/web/2019129014439/https://ouraring.com/tech-specs/>>, 3 pages Jan. 29, 2019.
- "Learn how the Oura ring works | Go inside | Oura Ring", available online at <<https://web.archive.org/web/20181127193557/https://ouraring.com/how-oura-works/>>, 5 pages Nov. 27, 2018.
- Oura Ring Generation 1 User Manual, available online at <<https://fccid.io/2AD7V-OURARING15001/User-Manual/User-Manual-2844448.pdf>>, 8 pages Nov. 24, 2015.
- Oura Ring Generation 2 User Manual, available online at <<https://fccid.io/2AD7V-OURA1801/User-Manual/User-manual-v2-3856414.pdf>>, 13 pages Mar. 27, 2018.
- Oura ring. Improve sleep. Perform better. by OURA—Kickstarter, available online at <<https://web.archive.org/web/20160427015852/https://www.kickstarter.com/projects/oura/oura-ring-improve-sleep-perform-better/description>>, 27 pages Apr. 27, 2016.
- Oura Ring review—The Gadgeteer, available online at <<https://the-gadgeteer.com/2017/08/11/oura-ring-review/>> 16 pages Aug. 11, 2017.
- Ouraring.com JZ50-0112 user manual available with Gen 2 ring purchase, 2 pages.
- Important Information Please Read JZ50-0148 user manual available with gen 2 ring purchase, 4 pages May 13, 2019.
- Important Information Please Read JZ50-0149 user manual available with gen 2 ring purchase, 4 pages May 13, 2019.
- Oura Ring | Sleep Tracker and Smart Ring with a Heart Rate Monitor, available online at <<https://web.archive.org/web/20180709050831/https://ouraring.com/>>, 12 pages Jul. 9, 2018.
- Introducing the New Oura Ring Generation 3—The Pulse Blog, available online at <<https://ouraring.com/blog/oura-generation2-vs-generation3/>>, 3 pages Oct. 26, 2021.
- Oura Ring 2 Teardown: Inside the NBA's COVID-19-Detecting Smart Ring, available at <<https://www.youtube.com/watch?v=BwA1hmSVgVY>> (Screenshots) Jul. 9, 2020.
- Oura Ring 2 Teardown: Inside the NBA's COVID-19-Detecting Smart Ring, available at <<https://www.youtube.com/watch?v=BwA1hmSVgVY>> (Script) Jul. 9, 2020.
- Int'l Search Report and Written Opinion for PCT/IB2008/053982 Mar. 2009.
- Hill et al., (2024) The Best Smart Ring to Rule Them All Retrieved from: <https://www.wired.com/gallery/best-smart-rings/> 2024.
- Kaltenbach, F. (2004). Translucent Materials: Glass, Plastics, Metals 2004.
- International Search Report and Written Opinion—PCT/US2007/081119, International Search Authority—European Patent Office—Nov. 4, 2009. Nov. 2009.
- Zhu et al., (2011). Wearable Sensor-Based Hand Gesture and Daily Activity Recognition for Robot-Assisted Living. IEEE Transactions on Systems, Man, and Cybernetics, Part A. 41. 569-573. 10.1109/TSMCA.2010.2093883. 2011.
- Stables et al., (2022). Why the Oura Ring Gen 3 was our wearable of the year. Retrieved from: <https://www.wearable.com/wearable-tech/why-the-oura-ring-3-was-our-wearable-of-the-year-2022> 2022.
- Sun et al., (2013). Wireless Power Transfer for Medical Microsystems. 10.1007/978-1-4614-7702-0. 2013.
- U.S. Appl. No. 61/768,279 2013.
- Sleep Lab validation of a wellness ring detecting sleep patterns based on photoplethysmogram, actigraphy and body temperature published Feb. 9, 2016, to Kinnunen (Year: 2016) 2016.
- Rhee et al., (2000). Artifact-resistant, power-efficient design of finger-ring plethysmographic sensors. I. Design and analysis. 4. 2792-2795 vol. 4. 10.1109/IEMBS.2000.901443. 2000.
- Rhee et al., (2000). Artifact-resistant, power-efficient design of finger-ring plethysmographic sensors. II. Prototyping and benchmarking. 4. 2796-2799 vol. 4. 10.1109/IEMBS.2000.901444. 2000.
- Dynamic drinkware-type analysis for mestas.
- Liu et al., (2009). UWave: Accelerometer-based personalized gesture recognition and its applications. Pervasive and Mobile Computing. 5. 657-675. 10.1016/j.pmcj.2009.07.007. 2009.
- Trigueiros et al., (2019). A comparison of machine learning algorithms applied to hand gesture recognition. 2019.
- Castaneda et al., (2018), Int J Biosens Bioelectron. 2018, "A review on wearable photoplethysmography sensors and their potential future applications in health care"; 4(4):195-202. doi: 10.15406/ijbsbe.2018.04.00125. 2018.
- Mendelson et al., (2006). A Wearable Reflectance Pulse Oximeter for Remote Physiological Monitoring. Conference proceedings : . . . Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Conference. 1. 912-5. 10.1109/IEMBS.2006.260137. 2006.
- Smiley, S., Active RFID vs. Passive RFID: What's the Difference? https://www.atlasrfidstore.com/rfid-insider/active-rfid-vs-passive-rfid/?srsltid=AfmBOoqhNhYwPPUSENIXB8LarZMm3TVQ4ugn4nTNUhfpY-9yYC_j0wdm Mar. 2016.
- Lawton, G., (2022). Active vs. passive RFID tags: Which to choose: TechTarget. Retrieved from: <https://www.techtarget.com/searcherp/tip/Active-vs-passive-RFID-tags-Which-to-choose> Nov. 2022.
- Amma et al., (2010). Airwriting recognition using wearable motion sensors. ACM International Conference Proceeding Series. 10. 10.1145/1785455.1785465. 2010.
- Zhou et al., (2012). Analysis and Selection of Features for Gesture Recognition Based on a Micro Wearable Device. International Journal of Advanced Computer Science and Applications. 3. 10.14569/IJACSA.2012.030101. 2012.
- Rhee et al., (2001). Artifact-resistant power-efficient design of finger-ring plethysmographic sensors. IEEE transactions on biomedical engineering. 48. 795-805. 10.1109/10.930904. 2001.
- Ying et al., (2007). Automatic Step Detection in the Accelerometer Signal. 10.1007/978-3-540-70994-7_14. 2007.
- 100 Best Inventions of 2020 <https://time.com/collection/best-inventions-2020/> 2020.
- Clark, B. A., "Color in Sunglass Lenses *," Optometry and Vision Science 46 (1969): 825-840. 1969.

(56)

References Cited**OTHER PUBLICATIONS**

Rhee, Sokwoo. (2006). Design and analysis of artifact-resistive finger photoplethysmographic sensors for vital sign monitoring /. 2006.

Webster, J.G., Design of pulse oximeters Webster—Institute of Physics Publishing—2003 2003.

Park et al., (2011). E-gesture: A collaborative architecture for energy-efficient gesture recognition with hand-worn sensor and mobile devices. SenSys 2011—Proceedings of the 9th ACM Conference on Embedded Networked Sensor Systems. 359-360. 10.1145/1999995.2000034. 2011.

Teh et al., (2000). Embedding of electronics within thermoplastic polymers using injection moulding technique. Filtration Industry Analyst. 10-18. 10.1109/IEMT.2000.910703. 2000.

Sakai, Tadamoto. (1993). Encapsulation process for electronic devices using injection molding method. Advances in Polymer Technology. 12. 61-71. 10.1002/adv.1993.060120106. 1993.

Ardebili et al.—William Andrew—2019. Encapsulation technologies for electronic applications 2019.

Au et al., (2009). Episodic Sampling: Towards Energy-efficient Patient Monitoring with Wearable Sensors. Conference proceedings : . . . Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Conference. 2009. 6901-5. 10.1109/IEMBS.2009.5333615. 2009.

Silverman, A. (2002). Fifty Years of Glass-Making. Industrial & Engineering Chemistry. 18. 10.1021/ie50201a004. 2002.

Petropoulos et al., (2012). Flexible PCB-MEMS flow sensor. Procedia Engineering. 47. 236-239. 10.1016/j.proeng.2012.09.127. 2012.

Schlömer et al., (2008). Gesture Recognition with a Wii Controller. First publ. in: Proceedings of the 2nd International Conference on Tangible and Embedded Interaction 2008, Bonn, Germany, Feb. 18-20, 2008, pp. 11-14. 10.1145/1347390.1347395. 2008.

Rekimoto, J. (2001). GestureWrist and GesturePad: unobtrusive wearable interaction devices. International Symposium on Wearable Computers, Digest of Papers. 21-27. 10.1109/ISWC.2001.962092. 2001.

Guidelines to Enhancing the Heart-Rate Monitoring Performance of Biosensing Wearables <https://www.analog.com/en/resources/technical-articles/guidelines-to-enhancing-the-heart-rate-monitoring-performance-of-biosensing-wearables.html> 2019.

Krzyanowski, J. "How are PCB's made? A Beginner's Guide to the PCB Manufacturing Process" Retrieved from Knowledge zone <https://vectorbluehub.com/how-are-pcbs-made> 2023.

Ciofu, C., et al., (2013) Injection and Micro Injection of Polymeric . . . First edition of the International Scientific Conference Modern Technologies in Machine Manufacturing Technology TCMC ISSN 2067-3604, vol. V, No. 1/2013 https://modtech.ro/international-journal/vol5no12013/Ciofu_Ciprian_1.pdf 2013.

Ross, R.J. (2004). LCP injection molded packages—keys to JEDEC 1 performance. 1807-1811 vol. 2. 10.1109/ECTC.2004.1320364. 2004.

Murphy, K. (2012) Machine learning: a probabilistic perspective 2012.

Olofson et al., Machining of titanium alloys—Battelle Memorial Institute, Defense Metals Information Center—1965 1965.

Asada et al., "Mobile monitoring with wearable photoplethysmographic biosensors," in IEEE Engineering in Medicine and Biology Magazine, vol. 22, No. 3, pp. 28-40, May-Jun. 2003, doi: 10.1109/EMMB.2003.1213624 2003.

Chen et al., (2009). Monitoring Human Movements at Home Using Wearable Wireless Sensors. Engineering Faculty Presentations. 2009.

On the Heels of 1 Million Rings Sold, Oura Now Valued at \$2.55 Billion [https://www.businesswire.com/news/home/20220405006108/en/On-the-Heels-of-1-Million-Rings-Sold-Oura-Now-Valued-at-2.55-Billion#:~:text=SAN%20FRANCISCO%2D%2D\(BUSINESS%20WIRE,of%20selling%201%2C000%2C000%20Oura%20Rings](https://www.businesswire.com/news/home/20220405006108/en/On-the-Heels-of-1-Million-Rings-Sold-Oura-Now-Valued-at-2.55-Billion#:~:text=SAN%20FRANCISCO%2D%2D(BUSINESS%20WIRE,of%20selling%201%2C000%2C000%20Oura%20Rings) Apr. 2022.

Lister et al., (2018) Optical properties of human skin <https://www.spiedigitallibrary.org/journals/journal-of-biomedical-optics/volume17/issue-9/090901/Optical-properties-of-human-skin/10.1117/1.JBO.17.9.090901.full> 2018.

Uchino et al., (2000). Prediction of Optical Properties of Commercial Soda-Lime-Silicate Glasses Containing Iron. Journal of Non-Crystalline Solids. 261. 72-78. 10.1016/S0022-3093(99)00617-1. 2000.

Lawrence et al., (1989) "Pulse Oximetry" Anesthesiology: The journal of the American Society of Anesthesiologists, Inc. vol. 70 No. 1 1989.

König et al., (1998) "Reflectance Pulse Oximetry-Principles and obstetric application in the zurich system" Journal of Clinical Monitoring and Computing vol. 14 No. 6PG. 3-6 1998.

* cited by examiner

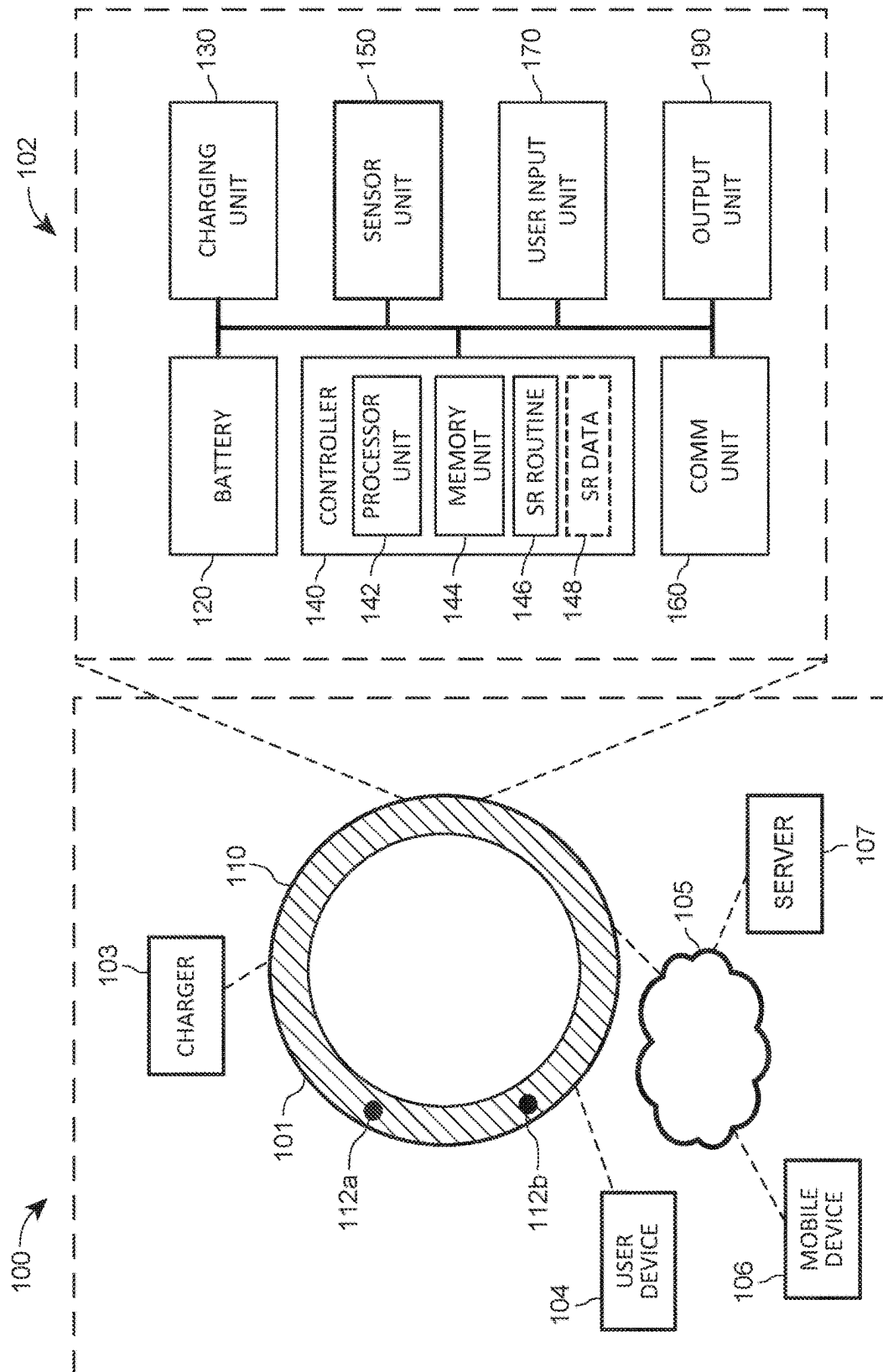


Figure 1

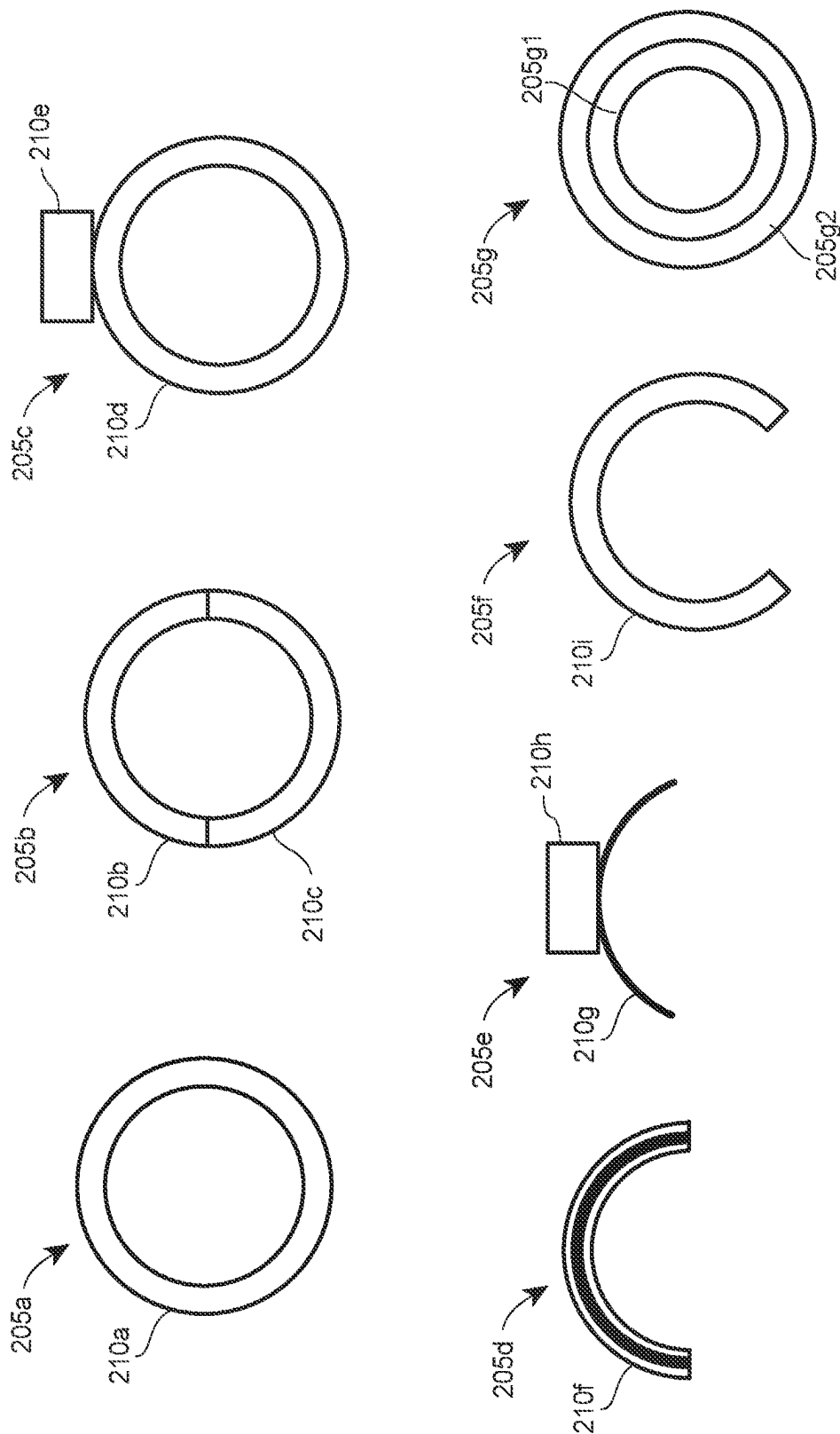


Figure 2

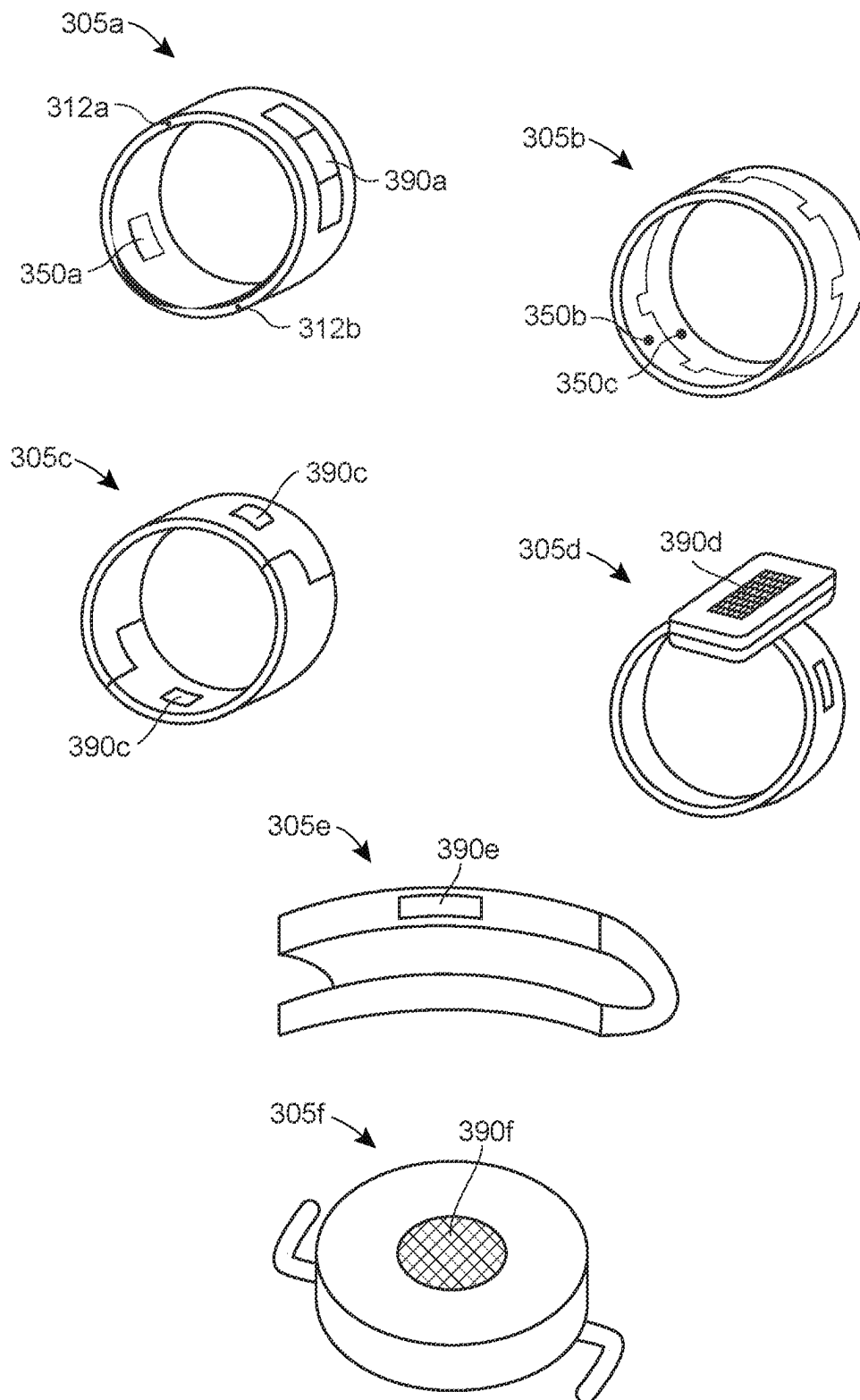


Figure 3

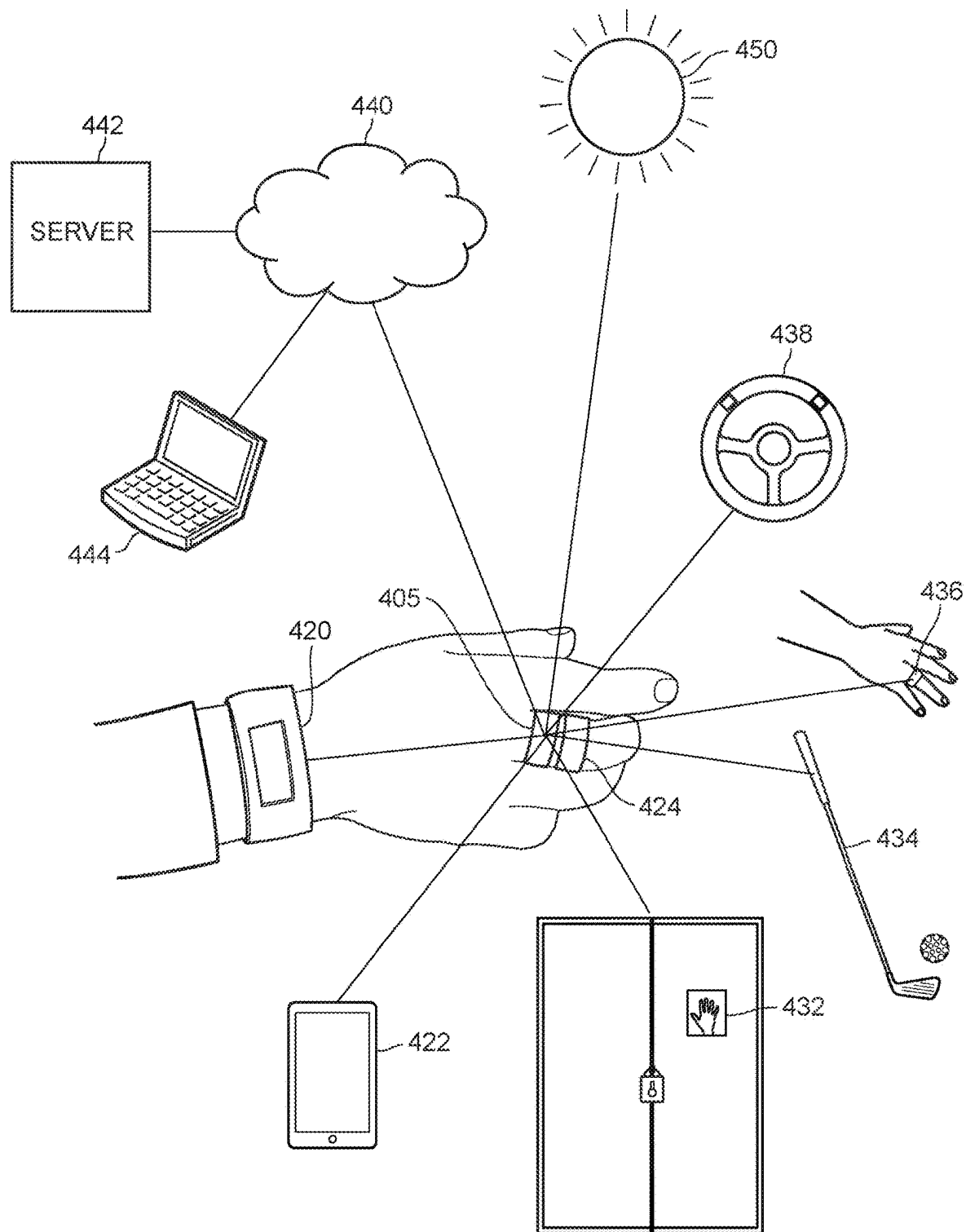


Figure 4

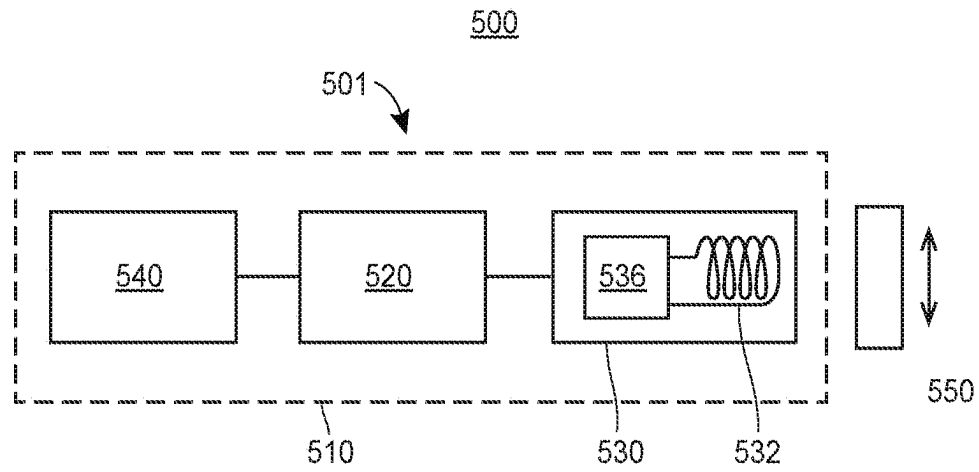


Figure 5

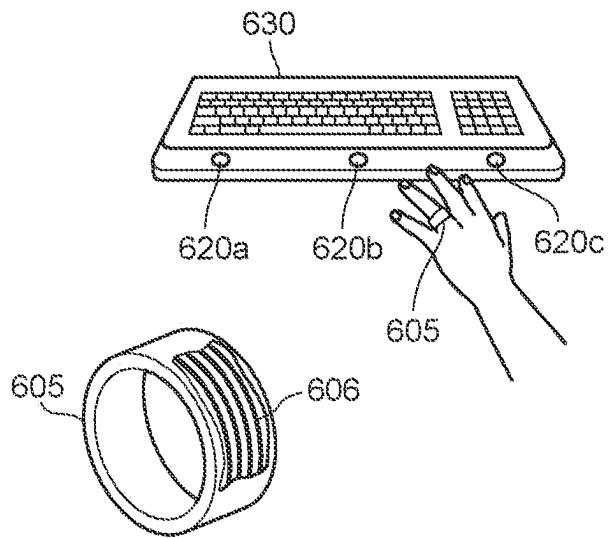


Figure 6A

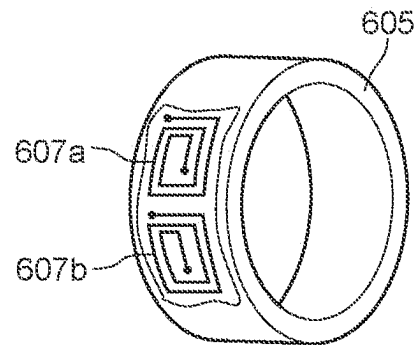


Figure 6B

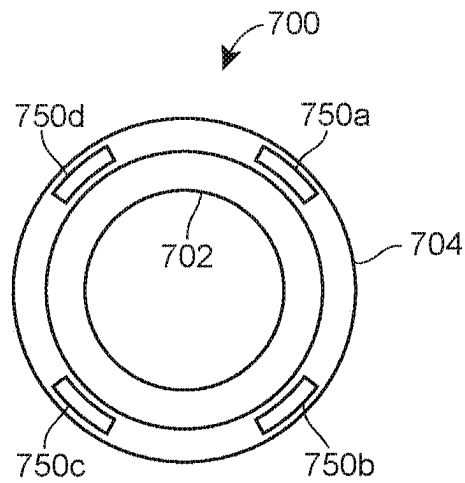


Figure 7A

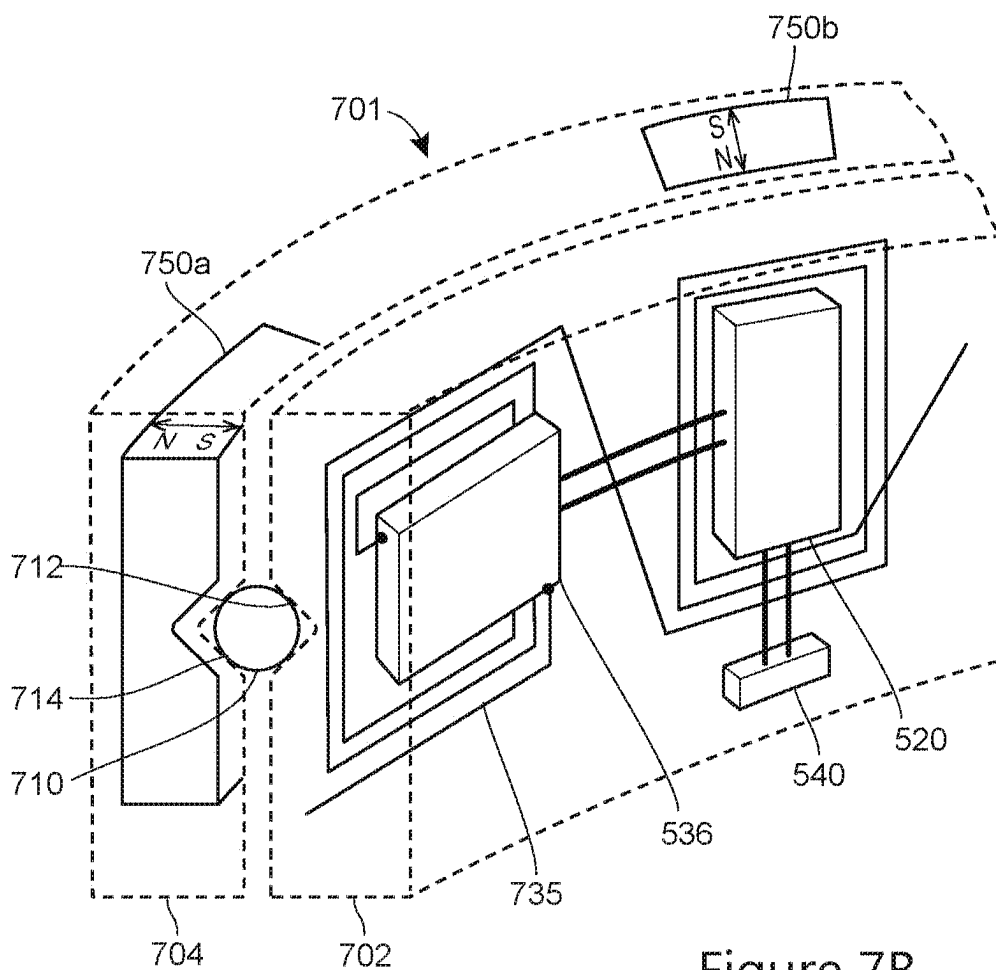


Figure 7B

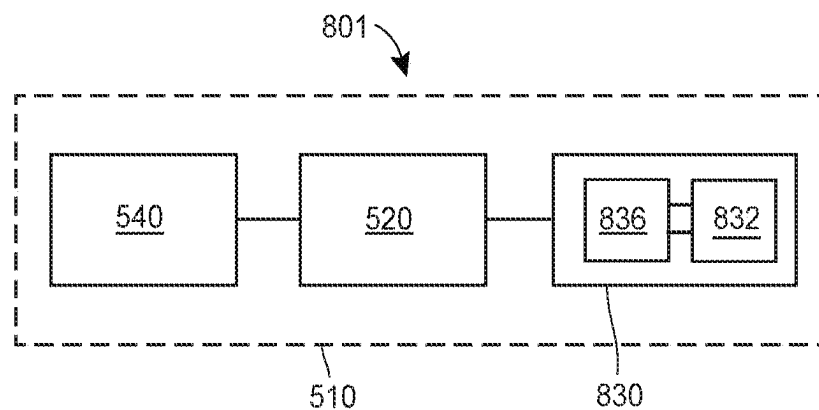


Figure 8

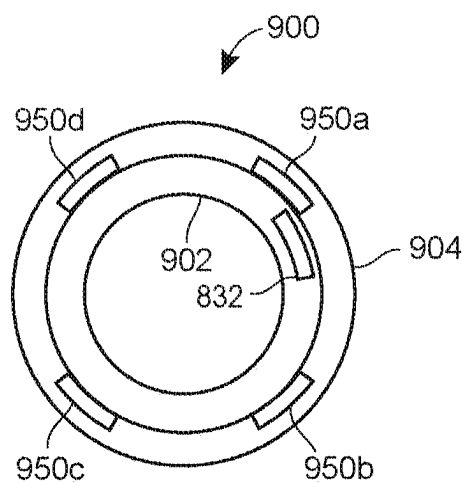


Figure 9A

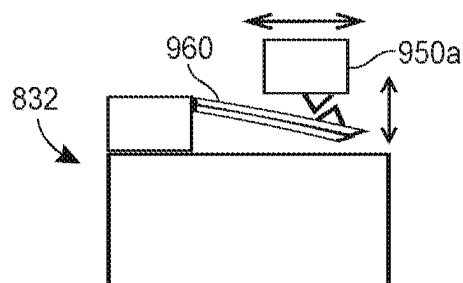


Figure 9B

1

HARVESTING ENERGY FOR A SMART RING VIA PIEZOELECTRIC CHARGING

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. Non-Provisional patent application Ser. No. 18/187,745, filed on Mar. 22, 2023, which claims priority to U.S. Non-Provisional Patent application Ser. No. 16/926,270 filed on Jul. 10, 2020, which claims priority to U.S. Provisional Patent Application No. 62/877,391, filed Jul. 23, 2019, and U.S. Provisional Patent Application No. 63/004,789, filed Apr. 3, 2020, all of which are incorporated by reference herein for all purposes.

FIELD OF THE DISCLOSURE

The present disclosure generally relates to implementations of smart ring wearable devices and, more specifically, to charging smart ring devices using energy harvested from motion.

BACKGROUND

To the extent that smart ring technology has been adopted, it has a number of challenges. For example, a number of problems exist with wearable devices generally, including: they often need to be removed for charging; they often have poor fit; they often provide relatively little user interactivity; and they often provide limited functionality.

BRIEF SUMMARY

A smart ring may be configured to harvest energy from user motion using a piezoelectric effect. To that end, the smart ring may include a piezoelectric harvesting element configured to generate a voltage in response to a deformation caused by motion.

In one aspect, a smart ring includes a ring-shaped housing, a power source disposed within the ring-shaped housing, and a charging circuit. The charging circuit includes a piezoelectric harvesting element, and is configured to charge the power source by way of electrical energy generated by mechanical deformation within the piezo-electric harvesting element. The smart ring further includes a component, disposed within the ring-shaped housing and configured to draw energy from the power source, and further configured to perform at least one of: i) sense a physical phenomenon external to the ring-shaped housing, ii) send communication signals to a communication device external to the ring-shaped housing, or iii) implement a user interface.

In another aspect, a method for operating a smart ring includes generating electrical energy at a charging circuit within the smart ring by way of a mechanical deformation in a piezoelectric harvesting element of the charging circuit. The method further includes charging a power source disposed within the ring-shaped housing of the smart ring by way of the charging circuit configured to deliver the generated electrical energy to the power source. Still further, the method includes operating a component, disposed within the ring-shaped housing of the smart ring and configured to draw energy from the power source, and further configured to perform at least one of: i) sense a physical phenomenon external to the ring-shaped housing, ii) send communication signals to a communication device external to the ring-shaped housing, or iii) implement a user interface.

2

In another embodiment, a smart ring includes a housing configured to be worn by a user. The smart ring can further include a power source disposed within the housing. The smart ring also can include a charging circuit disposed within the housing. The charging circuit can include a harvesting element configured to charge the power source. The harvesting element can be configured to convert mechanical energy to electrical energy. The smart ring also can include a device at least partially disposed within the housing. The device can be configured to draw energy from the power source.

In another embodiment, a smart ring includes a means for housing configured to be worn by a user. The smart ring can further include a means for powering disposed within the means for housing. The smart ring also can include a means for charging disposed within the means for housing. The means for charging can include a means for harvesting configured to charge the means for powering. The means for harvesting can be configured to convert mechanical energy to electrical energy. The smart ring also can include a means for drawing energy from the means for powering. The means for drawing energy from the means for powering can be at least partially disposed within the means for housing.

In another embodiment, a method for operating a smart ring includes generating electrical energy at a charging circuit within a smart ring. The charging circuit can include a harvesting element. The harvesting element can be configured to convert mechanical energy to electrical energy. The method for operating a smart ring further can include charging a power source disposed within the smart ring by delivering the electrical energy from the charging circuit to the power source. The method for operating a smart ring also can include causing the power source to provide power to a device disposed within the smart ring.

Depending upon the embodiment, one or more benefits may be achieved. These benefits and various additional objects, features and advantages of the present disclosure can be fully appreciated with reference to the detailed description and accompanying drawings that follow.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a system comprising a smart ring and a block diagram of smart ring components, according to some embodiments.

FIG. 2 illustrates a number of different form factor types of a smart ring, according to some embodiments.

FIG. 3 illustrates examples of different smart ring form factors.

FIG. 4 illustrates an environment within which a smart ring may operate, according to some embodiments.

FIG. 5 is a schematic diagram of a charging system for a smart ring, according to some embodiments.

FIG. 6A illustrates an example charging system that includes one or more permanent magnets disposed near or attached to a keyboard.

FIG. 6B illustrates a configuration of an induction coil for a charging system illustrated in FIG. 6A, according to some embodiments.

FIG. 7A illustrates an example charging system integrated into a smart ring that includes one or more permanent magnets disposed within a band of the smart ring.

FIG. 7B illustrates a section of the smart ring in FIG. 7A, according to some embodiments.

FIG. 8 is a schematic diagram of a smart ring configured to recharge by way of a piezoelectric harvesting element according to some embodiments.

3

FIG. 9A illustrates a smart ring configuration with two bands configured to move with respect to one another to generate mechanical energy for piezoelectric harvesting, according to some embodiments.

FIG. 9B illustrates a mechanism for causing deformation in a piezoelectric harvesting element when the bands in FIG. 9A move with respect to one another, according to some embodiments.

DETAILED DESCRIPTION

Smart ring wearable technology can enable a wide range of applications including security, safety, health and wellness, and convenient interfacing between a user and a variety of technologies based at least in part upon integrating a variety of sensor, input/output devices, and computing capabilities in a compact form factor. One of the challenges in increasing smart ring capabilities is reliably powering the needed components, particularly considering the limited space for a power source in the compact form factor. An ability to conveniently charge the power source of the smart ring without removing the smart ring from a finger would contribute to the adoption of smart ring technology.

One way to charge the smart ring without removing from the finger may include using a wearable charger, or a charger disposed in the environment of the smart ring. For the purposes of charging, the charger may be connected to the smart ring by a cable. Additionally or alternatively, the smart ring may include one or more circuits that enable the smart ring to harvest optical, thermal, mechanical and/or other sources of energy to charge and/or recharge the smart ring. Some techniques for harvesting mechanical energy may be based at least in part upon electromagnetic induction, or a piezoelectric effect, as described herein. Some techniques may rely on generated variable electromagnetic fields at a charging source that may couple a portion of the generated energy to a smart ring. Such configurations may be referred to as wireless charging, rather than harvesting configurations. A charging system may generate constant magnetic fields with constant current through a properly configured conductor. Such systems rely on availability of electrical energy to sustain a magnetic field. An electromagnetic induction charging system that uses permanent magnets has the advantage of obviating the need for using electrical energy to produce magnetic fields, on the other hand.

Various techniques, systems, and methods for charging a power source of a smart ring using electromagnetic induction are discussed below with reference to FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6A, FIG. 6B, FIG. 7A, FIG. 7B, FIG. 8, FIG. 9A, and FIG. 9B. In section I, a smart ring and inductive charging system is described with reference to FIG. 1. In section II, example smart ring form factor types and configurations to facilitate harvesting energy via induction are discussed with reference to FIG. 2 and FIG. 3. In section III, an example operating environment in which a smart ring may operate is described with reference to FIG. 4. In section IV, an example schematic for an induction charging system is described with reference to FIG. 5. In section V, example induction charging systems are described with reference to FIG. 6A, FIG. 6B, FIG. 7A, and FIG. 7B. In section VI, a smart ring configured for charging using a piezoelectric harvesting element is described with reference to FIG. 8, FIG. 9A, and FIG. 9B. In section VII, other considerations are described.

I. Examples of Smart Ring and Smart Ring Components

FIG. 1 illustrates a system 100 comprising a smart ring 101 that may be charged via an induction charging system

4

(e.g., by way of harvesting motion energy) according to one or more of the techniques described herein. FIG. 1 also shows one or more devices or systems that may be electrically, mechanically, or communicatively connected to the smart ring 101. As shown, the smart ring 101 may include a set of components 102, which may have various power needs and may impact the frequency with which the smart ring 101 needs recharging. To facilitate compatibility with the described energy harvesting techniques, the components 102 may be configured to be compatible with the magnetic fields and magnetic field variations to which the smart ring 101 and the components 102 may be exposed during induction charging. With respect to the described piezoelectric harvesting, the components 102 may be configured to be isolated from the potentially high voltages generated during piezoelectric harvesting.

The system 100 may comprise any one or more of: a charger 103 for the smart ring 101, a user device 104, a network 105, a mobile device 106, or a server 107. The charger 103 may provide energy to the smart ring 101 by way of a direct electrical, a wireless, or an optical connection. The smart ring 101 may be in a direct communicative connection with the user device 104, the mobile device 106, or the server 107 by way of the network 105. Interactions between the smart ring 101 and other components of the system 100 are discussed in more detail in the context of FIG. 4.

The smart ring 101 may sense a variety of signals indicative of activities of a user wearing the ring 101, biometric signals, a physiological state of the user, or signals indicative of the user's environment. The smart ring 101 may analyze the sensed signals using built-in computing capabilities or in cooperation with other computing devices (e.g., user device 104, mobile device 106, server 107) and provide feedback to the user or about the user via the smart ring 101 or other devices (e.g., user device 104, mobile device 106, server 107). Additionally or alternatively, the smart ring 101 may provide the user with notifications sent by other devices, enable secure access to locations or information, or a variety of other applications pertaining to health, wellness, productivity, or entertainment.

The smart ring 101, which may be referred to herein as the ring 101, may comprise a variety of mechanical, electrical, optical, or any other suitable subsystems, devices, components, or parts disposed within, at, throughout, or in mechanical connection to a housing 110 (which may be ring shaped and generally configured to be worn on a finger). Additionally, a set of interface components 112a and 112b may be disposed at the housing, and, in particular, through the surface of the housing. The interface components 112a and 112b may provide a physical access (e.g., electrical, fluidic, mechanical, or optical) to the components disposed within the housing. The interface components 112a and 112b may exemplify surface elements disposed at the housing. As discussed below, some of the surface elements of the housing may also be parts of the smart ring components.

As shown in FIG. 1, the components 102 of the smart ring 101 may be distributed within, throughout, or on the housing 110. As discussed in the contexts of FIG. 2 and FIG. 3 below, the housing 110 may be configured in a variety of ways and include multiple parts. The smart ring components 102 may for example, be distributed among the different parts of the housing 110, as described below, and may include surface elements of the housing 110. The housing 110 may include mechanical, electrical, optical, or any other suitable subsystems, devices, components, or parts disposed within or in mechanical connection to the housing 110, including a

5

battery 120, a charging unit 130, a controller 140, a sensor system 150 comprising one or more sensors, a communications unit 160, a one or more user input devices 170, or a one or more output devices 190. Each of the components 120, 130, 140, 150, 160, 170, 190 may include one or more associated circuits, as well as packaging elements. The components 120, 130, 140, 150, 160, 170, 190 may be electrically or communicatively connected with each other (e.g., via one or more busses or links, power lines, etc.), and may cooperate to enable “smart” functionality described within this disclosure.

The battery 120 may supply energy or power to the controller 140, the sensors 150, the communications unit 160, the user input devices 170, or the output devices 190. In some scenarios or implementations, the battery 120 may supply energy or power to the charging unit 130. The charging unit 130, may supply energy or power to the battery 120. In some implementations, the charging unit 130 may supply (e.g., from the charger 103, or harvested from other sources) energy or power to the controller 140, the sensors 150, the communications unit 160, the user input devices 170, or the output devices 190. In a charging mode of operation of the smart ring 101, the average power supplied by the charging unit 130 to the battery 120 may exceed the average power supplied by the battery 120 to the charging unit 130, resulting in a net transfer of energy from the charging unit 130 to the battery 120. In a non-charging mode of operation, the charging unit 130 may on average, draw energy from the battery 120.

The battery 120 may include one or more cells that convert chemical, thermal, nuclear or another suitable form of energy into electrical energy to power other components or subsystems 140, 150, 160, 170, 190 of the smart ring 101. The battery 120 may include one or more alkaline, lithium, lithium-ion and or other suitable cells. The battery 120 may include two terminals that, in operation, maintain a substantially fixed voltage of 1.5, 3, 4.5, 6, 9, 12 V or any other suitable terminal voltage between them. When fully charged, the battery 120 may be capable of delivering to power-sinking components an amount of charge, referred to herein as “full charge,” without recharging. The full charge of the battery may be 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000, 2000, 5000 mAh or any other suitable charge that can be delivered to one or more power-consuming loads as electrical current.

The battery 120 may include a charge-storage device, such as, for example a capacitor or a super-capacitor. In some implementations discussed below, the battery 120 may be entirely composed of one or more capacitive or charge-storage elements. The charge storage device may be capable of delivering higher currents than the energy-conversion cells included in the battery 120. Furthermore, the charge storage device may maintain voltage available to the components or subsystems 130, 140, 150, 160, 170, 190 when one or more cells of the battery 120 are removed to be subsequently replaced by other cells.

The charging unit 130 may be configured to replenish the charge supplied by the battery 120 to power-sinking components or subsystems (e.g., one or more of subsystems 130, 140, 150, 160, 170, 190) or, more specifically, by their associated circuits. To replenish the battery charge, the charging unit 130 may convert one form of electrical energy into another form of electrical energy. More specifically, the charging unit 130 may convert alternating current (AC) to direct current (DC), may perform frequency conversions of current or voltage waveforms, or may convert energy stored in static electric fields or static magnetic fields into direct

6

current. Additionally or alternatively, the charging unit 130 may harvest energy from radiating or evanescent electromagnetic fields (including optical radiation) and convert it into the charge stored in the battery 120. Furthermore, the charging unit 130 may convert non-electrical energy into electrical energy. For example, the charging unit 130 may harvest energy from motion, or from thermal gradients.

The controller 140 may include a processor unit 142 and a memory unit 144. The processor unit 142 may include one or more processors, such as a microprocessor (μ P), a digital signal processor (DSP), a central processing unit (CPU), a graphical processing unit (GPU), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or any other suitable electronic processing components. Additionally or alternatively, the processor unit 142 may include photonic processing components.

The memory unit 144 may include one or more computer memory devices or components, such as one or more registers, RAM, ROM, EEPROM, or on-board flash memory. The memory unit 144 may use magnetic, optical, electronic, spintronic, or any other suitable storage technology. In some implementations, at least some of the functionality the memory unit 144 may be integrated in an ASIC or and FPGA. Furthermore, the memory unit 144 may be integrated into the same chip as the processor unit 142 and the chip, in some implementations, may be an ASIC or an FPGA.

The memory unit 144 may store a smart ring (SR) routine 146 with a set of instructions, that, when executed by the processor 142 may enable the operation and the functionality described in more detail below. Furthermore, the memory unit 144 may store smart ring (SR) data 148, which may include (i) input data used by one or more of the components 102 (e.g., by the controller when implementing the SR routine 146) or (ii) output data generated by one or more of the components 102 (e.g., the controller 140, the sensor unit 150, the communication unit 160, or the user input unit 170). In some implementations, other units, components, or devices may generate data (e.g., diagnostic data) for storing in the memory unit 144.

The processing unit 142 may draw power from the battery 120 (or directly from the charging unit 130) to read from the memory unit 144 and to execute instructions contained in the smart ring routine 146. Likewise, the memory unit 144 may draw power from the battery 120 (or directly from the charging unit 130) to maintain the stored data or to enable reading or writing data into the memory unit 144. The processor unit 142, the memory unit 144, or the controller 140 as a whole may be capable of operating in one or more low-power mode. One such low power mode may maintain the machine state of the controller 140 when less than a threshold power is available from the battery 120 or during a charging operation in which one or more battery cells are exchanged.

The controller 140 may receive and process data from the sensors 150, the communications unit 160, or the user input devices 170. The controller 140 may perform computations to generate new data, signals, or information. The controller 140 may send data from the memory unit 144 or the generated data to the communication unit 160 or the output devices 190. The electrical signals or waveforms generated by the controller 140 may include digital or analog signals or waveforms. The controller 140 may include electrical or electronic circuits for detecting, transforming (e.g., linearly or non-linearly filtering, amplifying, attenuating), or con-

verting (e.g., digital to analog, analog to digital, rectifying, changing frequency) of analog or digital electrical signals or waveforms.

The sensor unit **150** may include one or more sensors disposed within or throughout the housing **110** of the ring **101**. Each of the one or more sensors may transduce one or more of: light, sound, acceleration, translational or rotational movement, strain, temperature, chemical composition, surface conductivity or other suitable signals into electrical or electronic sensors or signals. A sensor may be acoustic, photonic, micro-electro-mechanical systems (MEMS) sensors, chemical, micro-fluidic (e.g., flow sensor), or any other suitable type of sensor. The sensor unit **150** may include, for example, an inertial motion unit (IMU) for detecting orientation and movement of the ring **101**.

The communication unit **160** may facilitate wired or wireless communication between the ring **101** and one or more other devices. The communication unit **160** may include, for example, a network adaptor to connect to a computer network, and, via the network, to network-connected devices. The computer network may be the Internet or another type of suitable network (e.g., a personal area network (PAN), a local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), a mobile, a wired or wireless network, a private network, a virtual private network, etc.). The communication unit **160** may use one or more wireless protocols, standards, or technologies for communication, such as Wi-Fi, near field communication (NFC), Bluetooth, or Bluetooth low energy (BLE). Additionally or alternatively, the communication unit **160** may enable free-space optical or acoustic links. In some implementations, the communication unit **160** may include one or more ports for a wired communication connections. The wired connections used by the wireless communication module **160** may include electrical or optical connections (e.g., fiber-optic, twisted-pair, coaxial cable).

User input unit **170** may collect information from a person wearing the ring **101** or another user, capable of interacting with the ring **101**. In some implementations, one or more of the sensors in the sensor unit **150** may act as user input devices within the user input unit **170**. User input devices may transduce tactile, acoustic, video, gesture, or any other suitable user input into digital or analog electrical signal, and send these electrical signals to the controller **140**.

The output unit **190** may include one or more devices to output information to a user of the ring **101**. The one or more output devices may include acoustic devices (e.g., speaker, ultrasonic); haptic (thermal, electrical) devices; electronic displays for optical output, such as an organic light emitting device (OLED) display, a laser unit, a high-power light-emitting device (LED), etc.; or any other suitable types of devices. For example, the output unit **190** may include a projector that projects an image onto a suitable surface. In some implementations, the sensor unit **150**, the user input unit **170**, and the output unit **190** may cooperate to create a user interface with capabilities (e.g., a keyboard) of much larger computer systems, as described in more detail below.

The components **120**, **130**, **140**, **150**, **160**, **170**, **190** may be interconnected by a bus **195**, which may be implemented using one or more circuit board traces, wires, or other electrical, optoelectronic, or optical connections. The bus **195** may be a collection of electrical power or communicative interconnections. The communicative interconnections may be configured to carry signals that conform to any one

or more of a variety of protocols, such as I2C, SPI, or other logic to enable cooperation of the various components.

II. Example Smart Ring Form Factor Types

FIG. 2 includes block diagrams of a number of different example form factor types or configurations **205a**, **205b**, **205c** of a smart ring (e.g., the smart ring **101**). When the smart ring uses electromagnetic induction charging using permanent magnets, a form factor and configuration of the smart ring may influence or, conversely, depend on the implementation of a charging circuit. In particular, the configuration of an induction coil may be tied to a form factor of the smart ring, as discussed below.

The configurations **205a**, **205b**, **205c** (which may also be referred to as the smart rings **205a**, **205b**, **205c**) may each represent an implementation of the smart ring **101**, and each may include any one or more of the components **102** (or components similar to the components **102**). In some embodiments, one or more of the components **102** may not be included in the configurations **205a**, **205b**, **205c**. The configurations **205a**, **205b**, **205c** include housings **210a**, **210b**, **210c**, which may be similar to the housing **110** shown in FIG. 1.

The configuration **205a** may be referred to as a band-only configuration comprising a housing **210a**. In the configuration **205b**, a band may include two or more removably connected parts, such as the housing parts **210b** and **210c**. The two housing parts **210b** and **210c** may each house at least some of the components **102**, distributed between the housing parts **210b** and **210c** in any suitable manner.

The configuration **205c** may be referred to as a band-and-platform configuration comprising (i) a housing component **210d** and (ii) a housing component **210e** (sometimes called the “platform **210e**”), which may be in a fixed or removable mechanical connection with the housing **210d**. The platform **210e** may function as a mount for a “jewel” or for any other suitable attachment. The housing component **210d** and the platform **210e** may each house at least one or more of the components **102** (or similar components).

In some instances, the term “smart ring” may refer to a partial ring that houses one or more components (e.g., components **102**) that enable the smart ring functionality described herein. The configurations **205d** and **205e** may be characterized as “partial” smart rings, and may be configured for attachment to a second ring. The second ring may be a conventional ring without smart functionality, or may be second smart ring, wherein some smart functionality of the first or second rings may be enhanced by the attachment.

The configuration **205d**, for example, may include a housing **210f** with a groove to enable clipping onto a conventional ring. The grooved clip-on housing **210f** may house the smart ring components described above. The configuration **205e** may clip onto a conventional ring using a substantially flat clip **210g** part of the housing and contain the smart ring components in a platform **210h** part of the housing.

The configuration **205f**, on the other hand, may be configured to be capable of being mounted onto a finger of a user without additional support (e.g., another ring). To that end, the housing **210i** of the configuration **205f** may be substantially of a partial annular shape subtending between 180 and 360 degrees of a full circumference. When implemented as a partial annular shape, the housing **210i** may be more adaptable to fingers of different sizes than a fully annular band (360 degrees), and may be elastic. A restorative force produced by a deformation of the housing **210i** may

ensure a suitable physical contact with the finger. Additional suitable combinations of configurations (not illustrated) may combine at least some of the housing features discussed above.

The configuration **205g** may be configured to have two rings, a first ring **205g1** capable of and adapted to be mounted onto a finger of a user, and a second ring **205g2** capable of and adapted to be directly mounted onto the first ring **205g1**, as depicted in FIG. 2. Said another way, the first ring **205g1** and the second ring **205g2** are arranged in a concentric circle arrangement, such that the second ring **205g2** does not contact a user's finger when the smart ring **205g** is worn. Rather, only the first ring **205g1** contacts the user's finger. Additional suitable combinations of configurations (not illustrated) may combine at least some of the housing features discussed above.

FIG. 3 includes perspective views of example configurations **305a**, **305b**, **305c** of a smart ring (e.g., the smart ring **101**) in which a number of surface elements are included. When the smart ring uses electromagnetic induction charging using permanent magnets, a form factor and configuration of the smart ring may influence or, conversely, depend on the implementation of a charging circuit. In particular, the configuration of an induction coil may be tied to a form factor of the smart ring, as discussed below. Additionally, some of the surface elements may be configured to indicate when a magnetic field at the smart ring rises above or falls below a threshold.

Configuration **305a** is an example band configuration **205a** of a smart ring (e.g., smart ring **101**). Some of the surface elements of the housing may include interfaces **312a**, **312b** that may be electrically connected to, for example, the charging unit **130** or the communications unit **160**. On the outside of the configuration **305a**, the interfaces **312a**, **312b** may be electrically or optically connected with a charger to transfer energy from the charger to a battery (e.g., the battery **120**), or with another device to transfer data to or from the ring **305a**. The outer surface of the configuration **305a** may include a display **390a**, while the inner surface may include a biometric sensor **350a**.

The configurations **305b** and **305c** are examples of configurations of a smart ring with multiple housing parts (e.g., configuration **205b** in FIG. 2). Two (or more) parts may be separate axially (configuration **305b**), azimuthally (configuration **305c**), or radially (nested rings, not shown). The parts may be connected mechanically, electrically, or optically via, for example, interfaces analogous to interfaces **312a**, **312b** in configuration **305a**. Each part of a smart ring housing may have one or more surface elements, such as, for example, sensors **350b**, **350c** or output elements **390b**, **390c**. The latter may be LEDs (e.g., output element **390b**) or haptic feedback devices (e.g., output element **390c**), among other suitable sensor or output devices. Additionally or alternatively, at least some of the surface elements (e.g., microphones, touch sensors) may belong to the user input unit **170**.

Configuration **305d** may be an example of a band and platform configuration (e.g., configuration **205c**), while configurations **305e** and **305f** may be examples of the partial ring configurations **205d** and **205e**, respectively. Output devices **390d**, **390e**, **390f** on the corresponding configurations **305d**, **305e**, **305f** may be LCD display, OLED displays, e-ink displays, one or more LED pixels, speakers, or any other suitable output devices that may be a part of a suite of outputs represented by an output unit (e.g., output unit **190**). Other surface elements, such as an interface component **312c** may be disposed within, at, or through the housing. It

should be appreciated that a variety of suitable surface elements may be disposed at the illustrated configurations **305a**, **305b**, **305c** at largely interchangeable locations. For example, the output elements **390d**, **390e**, **390f** may be replaced with sensors (e.g., UV sensor, ambient light or noise sensors, etc.), user input devices (e.g., buttons, microphones, etc.), interfaces (e.g., including patch antennas or optoelectronic components communicatively connected to communications units), or other suitable surface elements.

III. Example Operating Environments for a Smart Ring

FIG. 4 illustrates an example environment **400** within which a smart ring **405** may be configured to operate. A portion of an electromagnetic induction charging system (e.g., one or more permanent magnets) may be disposed within the environment, as discussed below. In an embodiment, the smart ring **405** may be the smart ring **101**. In some embodiments, the smart ring **405** may be any suitable smart ring capable of providing at least some of the functionality described herein. Depending on the embodiment, the smart ring **405** may be configured in a manner similar or equivalent to any of the configurations **205a**, **205b**, **205c** or **305a**, **305b**, **305c** shown in FIG. 2 and FIG. 3.

The smart ring **405** may interact (e.g., by sensing, sending data, receiving data, receiving energy) with a variety of devices, such as bracelet **420** or another suitable wearable device, a mobile device **422** (e.g., a smart phone, a tablet, etc.) that may be, for example, the user device **104**, another ring **424** (e.g., another smart ring, a charger for the smart ring **405**, etc.), a secure access panel **432**, a golf club **434** (or another recreational accessory), a smart ring **436** worn by another user, or a steering wheel **438** (or another vehicle interface). Additionally or alternatively, the smart ring **405** may be communicatively connected to a network **440** (e.g., WiFi, 5G cellular), and by way of the network **440** (e.g., network **105** in FIG. 1) to a server **442** (e.g., server **107** in FIG. 1) or a personal computer **444** (e.g., mobile device **106**). Additionally or alternatively, the ring **405** may be configured to sense or harvest energy from natural environment, such as the sun **450**.

The ring **405** may exchange data with other devices by communicatively connecting to the other devices using, for example, the communication unit **160**. The communicative connection to other device may be initiated by the ring **405** in response to user input via the user input unit **170**, in response to detecting trigger conditions using the sensor unit **150**, or may be initiated by the other devices. The communicative connection may be wireless, wired electrical connection, or optical. In some implementation, establishing a communicative link may include establishing a mechanical connection.

The ring **405** may connect to other devices (e.g., a device with the charger **103** built in) to charge the battery **120**. The connection to other devices for charging may enable the ring **405** to be recharged without the need for removing the ring **405** from the finger. For example, the bracelet **420** may include an energy source that may transfer the energy from the energy source to the battery **120** of the ring **405** via the charging unit **430**. To that end, an electrical (or optical) cable may extend from the bracelet **420** to an interface (e.g., interfaces **112a**, **112b**, **312a**, **312b**) disposed at the housing (e.g., housings **110**, **210a**, **210b**, **210c**, **210d**, **210e**, **210f**, **210g**, **210h**, **210i**) of the ring **405**. The mobile device **422**, the ring **424**, the golf club **434**, the steering wheel **438** may also include energy source configured as chargers (e.g., the

charger 103) for the ring 405. The chargers may transfer energy to the ring 405 via a wired or wireless (e.g., inductive coupling) connection with the charging unit 130 of the ring 405.

In some implementations, the environment 400 may include elements to facilitate harvesting of energy produced by motion of a user wearing the ring 405 (e.g., an embodiment of the ring 101). For example, one or more permanent magnets may be disposed within the environment 400 (e.g., in the golf club 434, the steering wheel 438, or any other suitable location), configuring a magnetic field that the ring 405 may use for charging the battery 120.

The ring 405 may include (e.g., as a part of the charging unit 130) a charging circuit that, in turn, includes an induction coil. When user motion changes magnetic flux from the one or more permanent magnets through the induction coil, a current may be induced in the coil. The charging circuit may be configured to charge a power source (e.g., the battery 120) disposed within the ring 405 using the current induced in the coil.

IV. Example Diagram of an Electromagnetic Induction Charging System for a Smart Ring

FIG. 5 is a schematic diagram of a charging system 500 for a smart ring 501 (e.g., smart ring 101, smart ring 405) that may operate, for example, within the environment 400. The structure of the charging system 500 provides a general framework for implementations discussed in reference to FIG. 6A, FIG. 6B, FIG. 7A, FIG. 7B.

Disposed within a housing 510 of the smart ring, may be a power source 520 (e.g., battery 120), a charging circuit 530 (e.g., included in the charging unit 130) that includes an induction coil 532 and rectification components 536, and a smart-ring component 540 (referred to as component 540). The component 540 may be a part or an entirety of the controller 140, the sensor unit 150, the user input unit 170, the communications unit 160 or the output unit 190. The power source 520 may be in electrical connection with the charging circuit 530 and with the component 540. The charging circuit 530 may be configured to charge (add energy to) the power source 520, while the component 540 may be configured draw energy from the power source 520. The component 540 may be further configured to perform at least one of: i) sense a physical phenomenon external to the housing 510, ii) send communication signals to a communication device external to the housing 510, or iii) implement a user interface, as discussed above. The system 500 also includes one or more permanent magnets disposed in a magnet module 550. The magnet module 550 may be any aspect of the environment (e.g., in the golf club 434, the steering wheel 438 of the environment 400, in a keyboard of a computer, etc.) where one or more permanent magnets may be disposed to set up a magnetic field at the location of the ring. In certain examples, a magnet module 550 may be a part of the ring, as described in more detail below, in reference to FIG. 7. Changing a relative position of the coil 532 and the magnet module 550 may change magnetic flux from the one or more permanent magnets of the module 550 through the induction coil 532, generating electricity for charging the power source 520. Changing the relative position of the coil 532 and the magnet module 550 may be the result of motion of a user wearing the ring 501, for example.

The induction coil 532 may have 1, 3, 10, 30, 100, 300, 1000, 3000, 10000 or any other suitable number of turns of a conductive wire (or another suitable conductor, such as a trace). The conductive wire may have a diameter of 10 μ m,

20 μ m, 50 μ m, 100 μ m, 200 μ m, 500 μ m, 1 mm, 2 mm, or any other suitable diameter. The conductive wire may be at least in part made of copper, gold, or any other suitable conductive material and may include polymer coating to electrically isolate neighboring turns. The permanent magnets of the magnet module 550 may be rare earth magnets (e.g., neodymium, samarium-cobalt), alnico magnets, or ceramic magnets of suitable dimensions.

The rectifying components 536 included in the charging circuit 530 may be configured to convert the generally alternating current (AC) waveform generated in the coil 532 to a direct current (DC) output suitable for charging the power source 520. Smoothing the voltage excursions in the generated AC (e.g., by converting to DC) may increase the energy efficiency of charging. The rectifying components 536 may include discrete or integrated electronic components configured as filters, diode-bridge rectifiers, H-bridge rectifiers, buck converters, boost converters, buck-boost converters, or any other suitable circuits. In some implementations, the rectifying components 536 may include a commutator.

In a sense, the rectifying components 536 may be thought to include an rectifier configured to rectify or convert (e.g., in cooperation with a capacitor or another suitable component) AC to DC, thereby facilitating AC/DC conversion for current flowing between the coil 532 of the charging circuit and the remaining portion of the charging circuit. Depending on design considerations and various constraints, the rectifier may include a commutator or a solid-state rectifier. The commutator carries the advantage of rectifying the AC voltage across coil terminals without introducing a significant voltage drop and associated energy loss. The solid-state rectifier has fewer mechanical components than a commutator and may simplify the ring design, but may introduce a more significant voltage drop and energy loss than the commutator when rectifying.

V. Examples of the Induction Charging System for a Smart Ring

FIG. 6A illustrates an example charging system 600 (an embodiment of the charging system 500) for a smart ring 605 (e.g., that may be ring 101, ring 405, or ring 501) that includes an induction coil 606. Advantageously, the charging system 600 may harvest the energy from the motion of the ring worn by a user while typing. The charging system 600 includes permanent magnets 620a, 620b, 620c, disposed in a wrist rest 625 which is disposed near or attached to a keyboard 630.

The induction coil 606 may have 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000 or any other suitable number of turns of a conductive wire. The conductive wire may have a diameter of 10 μ m, 20 μ m, 50 μ m, 100 μ m, 200 μ m, 500 μ m, 1 mm, 2 mm, or any other suitable diameter. The conductive wire may be at least in part made of copper, gold, or any other suitable conductive material and may include polymer coating to electrically isolate neighboring turns.

The permanent magnets 620a, 620b, 620c may be rare earth magnets (e.g., neodymium, samarium-cobalt), alnico magnets, ceramic magnets. Though FIG. 6A illustrates three magnets 620a, 620b, 620c, the charging system 600 may use 1, 2, 5, 10, 20 or any other suitable number of magnets. The magnets 620a, 620b, 620c may have any suitable shapes, including, for example, cylinder, bar, block, or horseshoe. The magnets 620a, 620b, 620c may have any suitable dimensions. For example, the magnets 620a, 620b, 620c may be cylinders and have diameters of 2, 5, 10, 20 mm and

13

heights of 2, 5, 10, 20 mm. The magnets **620a**, **620b**, **620c** may be oriented in any suitable direction so as to have the magnetic flux density (magnetic field) gradients substantially maximized in a vertical (z-direction), a lateral (x-direction), or any other suitable direction. Generally, the magnets **620a**, **620b**, **620c** may be disposed so as to substantially maximize magnetic flux variations through the coil **606** during expected movement patterns of the ring **605** worn by a user. At a working distance (e.g., 5 mm to 10 cm) from the magnets **620a**, **620b**, **620c**, the magnetic field may vary from 20, 50, 100, 200, 500, 1000, 2000, 5000 gauss (or another suitable strength) close to the magnets to less than one gauss away from the magnets **620a**, **620b**, **620c**.

The induction coil **606** may have the turns wound substantially along the circumference of the ring **605**, as illustrated in FIG. 6A, with the axis of the coil **606** aligning axially with respect to the ring **605**. An axis of the coil here may be defined as the direction substantially orthogonal to a conductive path (e.g., wire or trace) of the coil. In another implementation, illustrated in FIG. 6B, one or more coils **607a**, **607b** may be disposed along the ring **605** with the axes oriented substantially radially with respect to the ring **605**. Furthermore, the coils **607a**, **607b** may be implemented as wires or as conductive material (e.g., gold, copper, etc.) traces (e.g., rectangular spiral traces) of any suitable width (20, 50, 100, 200, 500 μm , etc.) on a substrate. In some implementations, the substrate may be flexible.

In general, coils **606** and **607a**, **607b** may electrically connect to a number of devices or components (e.g., rectifying components **536**) to form a charging circuit (e.g., the charging circuit **530**). The charging circuit may include one or more rectifiers (e.g., implemented with diodes) and filters (e.g., implemented with capacitors), as well as voltage regulators or DC/DC voltage converters. In implementations with multiple coils (e.g., coils **607a**, **607b**), each coil may be connected to a corresponding rectifier. The multiple coils may be connected in series or in parallel, depending on an implementation of the charging circuit.

Returning to FIG. 6A, user movement (e.g., typing on the keyboard) may change magnetic flux through the coil **606** by varying magnetic field strength in the vicinity of the ring **605**, or varying the orientation of the coil **606** with respect to the direction of the magnetic field. The charging circuit may then charge the power source (e.g., the power source **520**) using the electricity generated in the coil. In implementations with multiple coils (e.g., coils **607a**, **607b**), charging circuit may be configured to combine the electricity generated in each coil for charging the power source.

FIG. 7A illustrates an example implementation of the charging system **500** fully integrated into a self-charging smart ring **700**. The self-charging smart ring **700** may allow a user to charge the smart ring **700** by spinning a band, for example. The user may thereby fidget with the smart ring **700** recreationally while simultaneously charging the ring **700**. FIG. 7B illustrates a section **701** of the smart ring **700** and, in more detail (and with a different perspective), components disposed within and a mechanical configuration of the smart ring **700**. The ring **700** includes a first band **702** and a second band **704** configured to rotate substantially coaxially with respect to the first band **702**. The second band **704** may include magnets **750a**, **750b**, **750c**, **750d**. In a sense, the second band **704** may be configured as the magnet module **550**. The first band **702**, on the other hand, may be configured as the housing **510**. Thus, in a sense, the smart ring **700** may be thought of as an implementation of the charging system **500** of FIG. 5, where the magnet module **550** is movably attached to the ring housing **510**. The power

14

source **520**, a coil **735** (e.g., coil **532**), rectifying components **536**, the component **540** may be disposed in the first band **702**.

In some implementations, the magnets **750a**, **750b**, **750c**, **750d** may be distributed around the second band **704** at regular radial intervals with alternating dipole orientations in a substantially radial direction. That is, half of the magnets **750a**, **750b**, **750c**, **750d** (or any even number of magnets) may be oriented with the north pole facing the radially inner surface of the second band **704** and the other half of the magnets **750a**, **750b**, **750c**, **750d** may be oriented with the north pole facing the radially outer surface of the second band **704**. Alternating magnet orientation around the circumference of the second band **704** may enhance the changes in magnetic flux through the coil **735** when the bands **702**, **704** rotate with respect to each other.

In some implementations, the first band **702** and the second band **704** may be radially disposed with respect to each other. An inner radius of at least a portion of the first band **702** may be larger than an outer radius of at least a portion of the second band **704**, with the first band **702** and the second band **704** forming, respectively, an outer portion of the ring **700** and an inner portion of the ring **700**. In certain examples, the first band **702** and the second band **704**, may form, respectively, the inner and the outer portions of the ring **700**. Still in other implementations, the first band **702** and the second band **704** may be substantially axially disposed with respect to each other.

In some implementations, the ring **700** may be configured as a radial ball bearing. To that end, the first band **702** and the second band **704** are configured as races for the radial ball bearing, in mechanical contact with each other by way of a suitable number of balls (e.g., ball **710**). The balls may be made of plastic, glass, ceramic, metal or metal alloy (e.g., steel) or any other suitable material. The bands **702**, **704** may include grooves **712**, **714** for confining the balls. The ring **700** may include a cage, a shield or other suitable radial ball bearing components. In some implementations, the ring **700** may be configured as a radial roller bearing, with the bands **702**, **704** configured as races for rollers.

In operation, a user may apply a force to rotate the second band **704** with respect to the first band **702** (or, conversely, the first band **702** with respect to the second **704**), actuating a spin of the second band **704**. The ring **700** may be configured to reduce frictional losses (e.g., using bearings as discussed above), so as to remain spinning for a certain duration (e.g., several seconds or tens of seconds) between manual actuations by the user. As the second band **704** spins with respect to the first band **702**, magnetic flux through the coil **735** changes due to the motion of the magnets **750a**, **750b**, **750c**, **750d** with respect to the coil **735**, inducing an AC voltage across coil terminals. The rectifying components **536** may then convert the electricity into DC form suitable for charging the power source **520**.

In some implementations, the rectifying components **536** may include a commutator. The commutator, for example, may mechanically reverse polarity of a connection between the coil **735** and the remaining portion of the charging circuit. The commutator may rectify the AC voltage across coil terminals without introducing a significant voltage drop and associated energy loss, as discussed above.

In other implementations, solid state rectification components may be used instead of the commutator to reduce the number of mechanical component and simplify the ring design, but may introduce a voltage drop and associated energy loss.

The second band **704** may be removably attached to the first band **702**. That is, removably attaching the second band **704** to the first band **702** may be done for the purpose of charging the smart ring **700**. Removably attaching here means that the user may attach or detach the second band **704** from the first band **702**. The detached second band may be removed from the finger. Removably attaching the second band **704** may preserve the rotational freedom of the second band **704** with respect to the first band. To that end, the second band **704** may include bearing balls, for example. The first band **702** may be worn on the finger of the user without the second band **704** to enable some or all of the functionality (e.g., other than charging) of the smart ring **700**. In some implementations the removable second band **704** may be configured as a radially outer ring, as described above. In other implementations, the removable second band may be disposed axially (e.g., along the finger of the user wearing the smart ring **700**) proximal to the first band **702** in a side-by-side configuration. In the side-by-side configuration, magnets **750a**, **750b**, **750c**, **750d** disposed in the second band may be configured to have magnetic dipoles aligned in the substantially axial direction. Correspondingly, the coil **735** in the first band **702** may be aligned to maximize magnetic flux and magnetic flux changes induced by the magnets **750a**, **750b**, **750c**, **750d**. In any case, the removable second band **704** may allow reducing the bulk of the smart ring during the segments of operation when charging is not needed.

VI. Examples of Piezoelectric Charging of a Smart Ring

Another technique for harvesting motion energy and converting it into electrical energy for use by a power source of a smart ring may include the use of a piezoelectric effect. The piezoelectric effect refers to the property of certain materials to generate voltage in response to mechanical stress that may produce microscopic deformations within the material. One advantage of using the piezoelectric effect is a relative ease of generating a substantial voltage (e.g., 1, 2, 5, 10, 20, 50, 100V or another suitable voltage), which may facilitate rectification.

Example implementations of a smart ring configured for piezoelectric harvesting of mechanical energy for charging are discussed below with reference to FIG. **8**, FIG. **9A**, and FIG. **9B**. FIG. **8** is a schematic diagram of a smart ring **801**, analogous to the smart ring **501** described in reference to FIG. **5**. The smart ring **801** may represent any one or more of the smart rings **205**, **305**, **405**, **605**, **700**, or **900**. The configuration of the ring **801** may have the housing **510**, the power source **520**, and the component **540** that is configured to draw energy from the power source **520**, as described above. A charging circuit **830** may replace the charging circuit **530** in FIG. **5**. The charging circuit **830** may be configured to charge the power source **520** by way of electrical energy generated by a piezoelectric harvesting element **832**, as discussed above. Referring to the FIG. **5**, the piezoelectric harvesting element **832** may take the place of the induction coil **532**. The charging circuit **830** may like the charging circuit **530**, include rectification components **836**.

The rectification components **836**, analogously to the rectification components **536** in FIG. **5**, may be configured to facilitate transforming the electrical energy generated by the piezoelectric harvesting element **832** into a form of electrical energy suitable for charging the power source **520**. Though the rectification components **836** may include similar elements to the rectification components **536**, e.g., diodes

(e.g., in a bridge configuration), capacitors, etc., the rectification components **836** may be configured differently. More specifically, the rectification components **836** may be configured for the electrical characteristics of the piezoelectric harvesting element **832**, that may have high output voltage and high output impedance. For example, the rectification components **836** may include a buck converter for stepping down the voltage output of the piezoelectric harvesting element and improving power transfer efficiency.

The piezoelectric harvesting element **832** may be disposed within the housing **510** of the smart ring **801** and configured for converting a mechanical energy input into electrical energy. The piezoelectric harvesting element **832** may be the only harvesting element in such a smart ring, or may be one of multiple harvesting elements. For example, the smart ring **801** may include the charging circuit **830** and the charging circuit **830** (which includes the piezoelectric harvesting element **832**) in an embodiment.

The piezoelectric harvesting element **832** may include a deformable structure (e.g., a cantilever, a membrane, or another suitable structure) made at least in part from a piezoelectric material, the structure configured to deform under an application of an external force. The macroscopic deformation of the structure in response to the applied force may result in microscopic deformations within the piezoelectric material of the structure, and, subsequently, in voltage generation. A cumulative effect of multiple mechanical deformations, whether cyclical or aperiodic, of the piezoelectric material or structure may generate a voltage waveform that may be converted (e.g., by rectification) to drive charge accumulation in the power source **520**.

After each mechanical deformation, the structure may substantially return to its original geometry, when the external force is no longer present, under the influence of the restorative force of the structure under deformation. In other implementations, the piezoelectric harvesting element may include an additional element (e.g., a spring) for providing a bias restorative force acting in opposition to the external force.

A variety of piezoelectric materials may be adapted for use within the piezoelectric harvesting element **832**. The materials may be crystalline or amorphous and may include ceramics (e.g., lead zirconate titanate known as PZT, sodium potassium niobate, etc.), semiconductors (e.g., gallium nitride, zinc oxide, etc.), and polymers (e.g., polyvinylidene fluoride, polyvinylidene fluoride, etc.). The materials may be grown, deposited, or otherwise fabricated into thin or thick films or any other suitable form. The deformable structure (e.g., membrane, cantilever, etc.) of the piezoelectric harvesting element may include one or more piezoelectric materials.

The deformable structure of the piezoelectric harvesting may be implemented as a microelectromechanical system (MEMS), using associated fabrication techniques. In other implementations, the deformable structure may be assembled from discrete mechanical components. Still in other implementations, the deformable structure may be implemented using additive manufacturing techniques, such as 3D printing. In certain examples, the deformable structure may be fabricated using a combination of suitable techniques.

In some implementations, the deformable structure of the piezoelectric harvesting element **832** may be configured to vibrate or oscillate in response to a force impulse. A natural, e.g., resonant, frequency of oscillation of the structure may depend, for example on the geometry and the distribution of mass of the deformable structure. The oscillation of the

deformable structure may induce a voltage oscillation at the corresponding frequency. The oscillation of the deformable structure may facilitate harvesting energy from periodic external forces, e.g., vibrations, particularly those with frequencies close to the natural or resonant frequency of the deformable structure. It may be advantageous to configure one or more deformable structures of the piezoelectric harvesting element **832** to have resonant frequencies close to expected frequencies of external forces or vibrations. The piezoelectric harvesting element **832** may be configured to have a resonant frequency in the range of vibration frequencies of a steering wheel or motorcycle handlebars, for example, to facilitate charging the smart ring **801** via piezoelectric energy harvesting while driving. Generally, configuring the deformable structure to have a resonant response to a vibration source may facilitate charging when the ring **801** is mechanically coupled to the vibration source. A configuration may include the frequency of the source within a 3 dB bandwidth of the resonant response of the deformable structure.

The piezoelectric harvesting element of the smart ring **801** may be configured to harvest energy from incidental movements of a user wearing the smart ring **801**. For example, the piezoelectric harvesting element may be configured to harvest energy from arm swings or finger tapping. To that end, a movable mass may be disposed within the housing **510** of the smart ring **801**. The movable mass may be mechanically coupled to the deformable structure of the piezoelectric harvesting element and exert a force on the deformable structure during acceleration associated with user movement and due to the inertia of the movable mass.

It may be noted that the deformable structure of the piezoelectric harvesting element may be configured with a degree of freedom in its movement aligned with any suitable direction with respect to the smart ring **801**. For example, the movement direction of the deformable element may be substantially radial with respect to the ring-shaped housing **510**. In certain examples, the degree of movement may be in a tangential direction, moving along the circumference of the smart ring **801**. In certain examples, the degree of movement may be aligned with the axial direction, or substantially along the finger of the user. Furthermore, the directions of movement may be combined, a deformable element may be configured for two- or three-axis deformations, or multiple structures may be configured with different degrees of freedom, to take advantage of a variety of movements.

In some implementations, the piezoelectric harvesting element may be configured to harvest energy from user movements that apply a force directly to the smart ring housing **510**, at least in part to charge the smart ring **801**. For example, the smart ring housing **510** may include a movable component that can translate a force, applied by the user, to the piezoelectric harvesting element. One suitable configuration may be analogous to the configuration of the smart ring discussed with reference to FIG. 7A and FIG. 7B. That is, in an example configuration, the smart ring **801** may include two bands configured to move with respect to each other to thereby generate mechanical energy for harvesting via the piezoelectric effect.

FIG. 9A and FIG. 9B illustrate a smart ring **900** that includes two bands **902**, **904** configured to rotate with respect to one another for the purpose of generating mechanical energy for piezoelectric harvesting. The smart ring **900** is analogous to the smart ring **700** in FIG. 7, in a sense that both allow a user to charge the smart rings **700**, **900** by rotating one of the movable bands **702**, **704**, **902**,

904. The ring **900** may include a first band **902**, in which the piezoelectric harvesting element **832** may be disposed, and a second band **904**. The first band **902** and the second band **904** may be configured, respectively, as an inner ring and an outer ring, where the outer ring is disposed substantially outside the inner ring in a radial direction relative to the inner ring. The smart ring **900** may be configured as a radial ball bearing, as discussed above for the smart ring **700**.

The second band **904** may include one or more actuation members **950a-d**, disposed so as to cause a mechanical deformation within the piezoelectric harvesting element **832** that, in turn, generates electrical energy for harvesting. As the first and second bands **902**, **904** move with respect to one another, the repeated mechanical deformation contributes to charging the power source **520**.

FIG. 9B illustrates how the actuation member **950a** causes the deformation in the piezoelectric harvesting element **832**. The piezoelectric harvesting element **832** may include a cantilever **960** that may have a protrusion or a catch that comes into contact with a corresponding protrusion on the actuation member **950a**, as the actuation member **950a** moves by the harvesting element **832**, the cantilever **960** deforms under the force applied by the actuation member **950a**.

The piezoelectric harvesting element may include one or more cantilevers (e.g., cantilever **960**) disposed along the circumference of the first band **902**. In some implementations, the one or more cantilevers may be oriented substantially in a tangential orientation with respect to the band **902**. In other implementations, the one or more cantilevers may be oriented radially with respect to the band **902**.

Though the illustrated mechanism relies on mechanical contact between the actuation member **950a** and the cantilever **960**, the piezoelectric harvesting element **832** may rely on other modes of actuation (e.g., magnetic, electrostatic). For example, the actuation member **950a** may be a magnet, and the cantilever **960** may include paramagnetic or ferromagnetic material that is repelled or attracted by a passing magnet. Thus, as the user rotates the second band **904**, the magnets may cause intermittent deformations in the cantilevers (e.g., cantilever **960**) or other deformable structures of the piezoelectric harvesting element **832**, causing the generation of electrical energy for charging the power source **520**.

It should be noted, that, though electromagnetic induction and piezoelectric effect are described in detail herein, other energy harvesting technologies may be integrated into a smart ring. Additional or alternative harvesting modules may include triboelectric, thermoelectric, or photovoltaic charging, for example.

VII. Examples of Other Considerations

When implemented in software, any of the applications, services, and engines described herein may be stored in any tangible, non-transitory computer readable memory such as on a magnetic disk, a laser disk, solid state memory device, molecular memory storage device, or other storage medium, in a RAM or ROM of a computer or processor, etc. Although the example systems disclosed herein are disclosed as including, among other components, software or firmware executed on hardware, it should be noted that such systems are merely illustrative and should not be considered as limiting. For example, it is contemplated that any or all of these hardware, software, and firmware components could be embodied exclusively in hardware, exclusively in software, or in any combination of hardware and software.

Accordingly, while the example systems described herein are described as being implemented in software executed on a processor of one or more computer devices, persons of ordinary skill in the art will readily appreciate that the examples provided are not the only way to implement such systems.

The described functions may be implemented, in whole or in part, by the devices, circuits, or routines of the system 100 shown in FIG. 1. Each of the described methods may be embodied by a set of circuits that are permanently or semi-permanently configured (e.g., an ASIC or FPGA) to perform logical functions of the respective method or that are at least temporarily configured (e.g., one or more processors and a set instructions or routines, representing the logical functions, saved to a memory) to perform the logical functions of the respective method.

Throughout this specification, some of the following terms and phrases are used.

Communication Interface according to some embodiments: Some of the described devices or systems include a “communication interface” (sometimes referred to as a “network interface”). A communication interface enables the system to send information to other systems and to receive information from other systems, and may include circuitry for wired or wireless communication.

Each described communication interface or communications unit (e.g., communications unit 160) may enable the device of which it is a part to connect to components or to other computing systems or servers via any suitable network, such as a personal area network (PAN), a local area network (LAN), or a wide area network (WAN). In particular, the communication unit 160 may include circuitry for wirelessly connecting the smart ring 101 to the user device 104 or the network 105 in accordance with protocols and standards for NFC (operating in the 13.56 MHz band), RFID (operating in frequency bands of 125-134 kHz, 13.56 MHz, or 856 MHz to 960 MHz), Bluetooth (operating in a band of 2.4 to 2.485 GHz), Wi-Fi Direct (operating in a band of 2.4 GHz or 5 GHz), or any other suitable communications protocol or standard that enables wireless communication.

Communication Link according to some embodiments: A “communication link” or “link” is a pathway or medium connecting two or more nodes. A link between two end-nodes may include one or more sublinks coupled together via one or more intermediary nodes. A link may be a physical link or a logical link. A physical link is the interface or medium(s) over which information is transferred, and may be wired or wireless in nature. Examples of physical links may include a cable with a conductor for transmission of electrical energy, a fiber optic connection for transmission of light, or a wireless electromagnetic signal that carries information via changes made to one or more properties of an electromagnetic wave(s).

A logical link between two or more nodes represents an abstraction of the underlying physical links or intermediary nodes connecting the two or more nodes. For example, two or more nodes may be logically coupled via a logical link. The logical link may be established via any combination of physical links and intermediary nodes (e.g., routers, switches, or other networking equipment).

A link is sometimes referred to as a “communication channel.” In a wireless communication system, the term “communication channel” (or just “channel”) generally refers to a particular frequency or frequency band. A carrier signal (or carrier wave) may be transmitted at the particular frequency or within the particular frequency band of the channel. In some instances, multiple signals may be trans-

mitted over a single band/channel. For example, signals may sometimes be simultaneously transmitted over a single band/channel via different sub-bands or sub-channels. As another example, signals may sometimes be transmitted via the same band by allocating time slots over which respective transmitters and receivers use the band in question.

Memory and Computer-Readable Media according to some embodiments: Generally speaking, as used herein the phrase “memory” or “memory device” refers to a system or device (e.g., the memory unit 144) including computer-readable media (“CRM”). “CRM” refers to a medium or media accessible by the relevant computing system for placing, keeping, or retrieving information (e.g., data, computer-readable instructions, program modules, applications, routines, etc.). Note, “CRM” refers to media that is non-transitory in nature, and does not refer to disembodied transitory signals, such as radio waves.

The CRM may be implemented in any technology, device, or group of devices included in the relevant computing system or in communication with the relevant computing system. The CRM may include volatile or nonvolatile media, and removable or non-removable media. The CRM may include, but is not limited to, RAM, ROM, EEPROM, flash memory, or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store information, and which can be accessed by the computing system. The CRM may be communicatively coupled to a system bus, enabling communication between the CRM and other systems or components coupled to the system bus. In some implementations the CRM may be coupled to the system bus via a memory interface (e.g., a memory controller). A memory interface is circuitry that manages the flow of data between the CRM and the system bus.

Network according to some embodiments: As used herein and unless otherwise specified, when used in the context of system(s) or device(s) that communicate information or data, the term “network” (e.g., the networks 105 and 440) refers to a collection of nodes (e.g., devices or systems capable of sending, receiving or forwarding information) and links which are connected to enable telecommunication between the nodes.

Each of the described networks may include dedicated routers responsible for directing traffic between nodes, and, in certain examples, dedicated devices responsible for configuring and managing the network. Some or all of the nodes may be also adapted to function as routers in order to direct traffic sent between other network devices. Network devices may be inter-connected in a wired or wireless manner, and network devices may have different routing and transfer capabilities. For example, dedicated routers may be capable of high volume transmissions while some nodes may be capable of sending and receiving relatively little traffic over the same period of time. Additionally, the connections between nodes on a network may have different throughput capabilities and different attenuation characteristics. A fiberoptic cable, for example, may be capable of providing a bandwidth several orders of magnitude higher than a wireless link because of the difference in the inherent physical limitations of the medium. If desired, each described network may include networks or sub-networks, such as a local area network (LAN) or a wide area network (WAN).

Node according to some embodiments: Generally speaking, the term “node” refers to a connection point, redistrib-

21

bution point, or a communication endpoint. A node may be any device or system (e.g., a computer system) capable of sending, receiving or forwarding information. For example, end-devices or end-systems that originate or ultimately receive a message are nodes. Intermediary devices that receive and forward the message (e.g., between two end-devices) are also generally considered to be “nodes.”

Processor according to some embodiments: The various operations of example methods described herein may be performed, at least partially, by one or more processors (e.g., the one or more processors in the processor unit 142). Generally speaking, the terms “processor” and “microprocessor” are used interchangeably, each referring to a computer processor configured to fetch and execute instructions stored to memory. By executing these instructions, the processor(s) can carry out various operations or functions defined by the instructions. The processor(s) may be temporarily configured (e.g., by instructions or software) or permanently configured to perform the relevant operations or functions (e.g., a processor for an Application Specific Integrated Circuit, or ASIC), depending on the particular embodiment. A processor may be part of a chipset, which may also include, for example, a memory controller or an I/O controller. A chipset is a collection of electronic components in an integrated circuit that is typically configured to provide I/O and memory management functions as well as a plurality of general purpose or special purpose registers, timers, etc. Generally speaking, one or more of the described processors may be communicatively coupled to other components (such as memory devices and I/O devices) via a system bus.

The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other embodiments the processors may be distributed across a number of locations.

Although specific embodiments of the present disclosure have been described, it will be understood by those of skill in the art that there are other embodiments that are equivalent to the described embodiments. Accordingly, it is to be understood that the present disclosure is not to be limited by the specific illustrated embodiments.

What is claimed is:

1. A smart ring comprising:

- a housing configured to be worn by a user;
- a power source disposed within the housing;
- a charging circuit disposed within the housing, wherein the charging circuit includes a harvesting element configured to charge the power source, and wherein the harvesting element is configured to harvest energy from a motion of a portion of the user wearing the housing; and
- a device at least partially disposed within the housing, wherein the device is configured to draw energy from the power source.

2. The smart ring of claim 1, wherein: the harvesting element comprises a piezoelectric harvesting element.

3. The smart ring of claim 1, wherein: the harvesting element comprises an induction coil.

4. The smart ring of claim 3, wherein: the induction coil includes one or more turns along a circumference of a ring-shaped housing.

22

5. The smart ring of claim 1, wherein:

the harvesting element includes one or more additively manufactured components.

6. The smart ring of claim 1, wherein:

the motion of the portion of the user comprises an arm swing.

7. The smart ring of claim 1, wherein:

the harvesting element is configured to vibrate or oscillate in response to the motion of the portion of the user wearing the housing.

8. A smart ring comprising:

a means for housing, wherein the means for housing is configured to be worn by a user;

a means for powering, wherein the means for powering is disposed within the means for housing;

a means for charging disposed within the means for housing, wherein the means for charging includes a means for harvesting configured to charge the means for powering, and wherein the means for harvesting is configured to harvest energy from a motion of a portion of the user wearing the housing; and

a means for drawing energy from the means for powering, wherein the means for drawing energy from the means for powering is at least partially disposed within the means for housing.

9. The smart ring of claim 8, wherein:

the means for harvesting is a means for harvesting piezoelectric energy.

10. The smart ring of claim 8, wherein:

the means for harvesting comprises an induction coil.

11. The smart ring of claim 10, wherein:

the induction coil includes one or more turns along a circumference of a ring-shaped housing.

12. The smart ring of claim 8, wherein:

the means for harvesting includes one or more additively manufactured components.

13. The smart ring of claim 8, wherein:

the motion of the portion of the user comprises an arm swing.

14. A method for operating a smart ring, the method comprising:

generating electrical energy at a charging circuit within the smart ring, wherein the charging circuit includes a harvesting element, and wherein the harvesting element is configured to harvest energy from a motion of a portion of a user wearing the smart ring;

charging a power source disposed within the smart ring by delivering the electrical energy from the charging circuit to the power source; and

causing the power source to provide power to a device disposed within the smart ring.

15. The method of claim 14, wherein:

the harvesting element comprises a piezoelectric harvesting element.

16. The method of claim 14, wherein:

the harvesting element comprises an induction coil.

17. The method of claim 16, wherein:

the induction coil includes one or more turns along a circumference of the smart ring.

18. The method of claim 14, wherein:

the harvesting element includes one or more additively manufactured components.

19. The method of claim 14, wherein:

the motion of the portion of the user comprises an arm swing.

23**24**

20. The method of claim 14, wherein:
the harvesting element is configured to vibrate or oscillate
in response to the motion of the portion of the user
wearing the smart ring.

* * * * *